UTHM CASE STUDY
STUDENT CLUBS AND ACTIVITIES MANAGEMENT

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CHAPTER 1
INTRODUCTION

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is a public university in Malaysia that supports a wide range of academic, administrative and student-related operations. Currently, UTHM implements various digital platforms Student Information System (SMP), Staff Information System (SMS), the UTHM SSO portal and MyUTHM mobile application to support daily operations.

As the university continues to expand its digital services, the need for integrated and efficient digital systems has become increasingly important. This project focuses on analysing UTHM's digital environment to propose an enterprise-level solution by applying Enterprise Architecture (EA) principles and developing a prototype using SAP Build Apps.

1.2 Problem Statement

Despite the availability of multiple digital systems at UTHM, many operational areas still operate independently with limited integration. This results in data fragmentation, inconsistent workflow across academic and administrative functions. Besides, users need to frequently switch between multiple platforms and external websites to complete their tasks, which can lead to inefficiencies.

Furthermore, several processes remain manual and the absence of a comprehensive Enterprise Architecture restricts the university to gain a holistic view of system interactions and data flow. This situation can significantly affect the decision making and future improvement of the university's digital environment.

1.3 Project Objectives

The objectives of the project are:

(a) To analyze UTHM's existing enterprise system and identify weaknesses in current business processes and system integration

(b) To design an enterprise-level system using Enterprise Architecture (EA) principles

(c) To develop and implement enterprise solutions using SAP Build Apps to improve data management and system integration.

1.4 Project Scope

The project focuses on the analysis, design, and development of an enterprise subsystem within Universiti Tun Hussein Onn Malaysia (UTHM) as a case study organization. The scope of this project is defined as follows:

(a) The project will be conducted within UTHM's digital ecosystem, involving students, academic staff and administrative staff.

(b) The system analysis will focus on UTHM's existing digital platforms, including Student Information System (SMP), Staff Information System (SMS), UTHM Single Sign-On (SSO), and the MyUTHM mobile application.

(c) The project will focus on the development of Enterprise Architecture (EA) to show the business, application, data and technology layers of UTHM's digital ecosystem integration.

(d) The system implementation will be limited to selected sub-system only and developed as a prototype using SAP Build Apps.

(e) The system will simulate integration with existing UTHM master data and workflows.

1.5 Project Importance

The project is important as it demonstrates the implementation of Enterprise Architecture (EA) into a higher education institution to achieve an improvement in data management, system integration and operational efficiency. The proposed solution provides a structural approach to manage the business processes, applications, data and technology within UTHM's digital ecosystem. This benefits both users and the university management team by providing an effective workflow and improving accessibility through a more integrated digital environment.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter reviews relevant literature related to Enterprise Architecture (EA) and its application in higher education institutions, with a focus on university enterprise systems. It also discusses studies related to Student Clubs and Activities Management Systems, highlighting existing approaches, challenges and best practices. In addition, this chapter reviews the technologies used in developing enterprise-level subsystems, particularly low-code platforms and integrated databases. The findings from this literature review provide theoretical and technical support for the development of the proposed Student Clubs & Activities Management System for Universiti Tun Hussein Onn Malaysia (UTHM).

2.2 Enterprise Architecture

This section provides the definition and concept of enterprise architecture highlighting its purpose and main domains in the architecture. The review of previous studies related to enterprise architecture (university enterprise architecture) are also discussed in this section.

2.2.1 Definition and Concept of Enterprise Architecture

Enterprise architecture (EA) is a well-defined approach or practice consisting of principles, methods and models that are used to design and manage the enterprise's core components which are organizational structure, business processes, information systems, and infrastructure (Jonkers et al., 2006). The purpose of EA is to ensure the alignment and integration between business and IT solutions to support the enterprise to achieve their objective in a more unified approach (Iyamu, 2022).

Enterprise architecture consists of four primary layer domains, namely business architecture, application architecture, data architecture, and technology architecture. Each domain has its own distinctive function, and all domains are interrelated with one another in supporting organizational strategy and operations (Iyamu, 2022). The business architecture domain provides the mission, goals, and objectives of the organization, which serve as key components that guide the development and implementation of the other domains (Iyamu, 2022). The application architecture domain focuses on the development and management of business and system applications within the organization (Iyamu, 2022). This domain ensures the integration and coexistence of applications within the system based on established principles and policies (Iyamu, 2022; Ulmi et al., 2020).

The data architecture domain is primarily responsible for the management of organizational data. It provides the structural design and approach for managing organizational information to support the optimal and efficient execution of enterprise processes and activities (Iyamu, 2022; Ulmi et al., 2020). Finally, the technology architecture domain focuses on the technological capabilities required to enable and support the organization in achieving its goals and objectives (Iyamu, 2022; Ulmi et al., 2020). EA frameworks such as TOGAF are developed to provide structured and high-level guidelines to develop and implement the EA (Jonkers et al., 2006; Ulmi et al., 2020). Among the various benefits of EA, one of the most significant is the improved integration of the organizational process and workflow allowing the organization to have better governance and workflow control (Iyamu, 2022; Ulmi et al., 2020).

2.2.2 Review of Previous Studies Related to Enterprise Architecture

To address the increasing system complexity and rapid growth of data within the university's system, several studies have explored the implementation of Enterprise Architecture as an approach to ensuring systemic and seamless integration. For instance, a study conducted in Bandung highlighted that a university traditionally manages their business processes and IT systems independently. This isolated management leads to increased complexity over time, ultimately hindering effective strategic planning and organizational alignment (Amalia & Supriadi, 2017).

The practical implementation of Enterprise Architecture in universities typically follows established frameworks with a significant emphasis on phased development cycles. Implementation begins by identifying and classifying the university's business processes into four core layers. The most commonly used of enterprise architecture frameworks is The Open Group Architecture Framework (TOGAF). In which, EA frameworks can provide a strategic method to design and manage the Information Technology (IT) and business process in enterprises such as universities (Meligy, 2025). To illustrate, the Business Architecture layer includes the primary goals and objectives of the university such as enhancing the learning and researching process for the university users (Harwikarya & Yusuf, 2016).

Then, the next 3 layers are designed strategically to allow the university to achieve its primary goal and objectives. For example, Harwikarya and Yusuf (2016) explain that the data layer is focused on how the data is stored and used to achieve the primary goals to allow seamless data usage and ensure a suitable model for communication with the stakeholders. Harwikarya and Yusuf (2016) also explains the application layer is to define necessary applications to manage data and provide information for the execution of business processes in university. In the last layer, technology architecture, this layer concerns the technology used for the system such as the hardware aspect to support the academic processes and ensure the adaptability, interoperability and integration of each layer (Harwikarya & Yusuf, 2016).

Previous studies have stated that the benefits of implementing EA in universities are improving business processes alignment along with the technology to ensure smooth integration and communication between those two domains (Harwikarya & Yusuf, 2016). Besides that, implementation of EA also can increase the operational efficiency in which business processes are streamlined, reducing manual and redundant tasks (Iswahyudi et al., 2023). However, the previous studies also stated important challenges in EA implementation which are existing system complexity where IT infrastructure and each business process are managed independently making full integration difficult (Ulmi et al., 2020). Furthermore, universities frequently encounter resistance to structural changes making it difficult to align legacy systems to new adaptive architectural models (Amalia & Supriadi, 2017).

Ultimately, this project is grounded in these findings. By adopting the principles of EA at a systemic level, this research seeks to demonstrate how a university can overcome isolated management patterns. The proposed architecture serves as a blueprint to ensure that digital initiatives are not merely standalone tools but are strategically integrated components that support the university's long-term vision of digital transformation.

2.3 Related Sub-System

This section describes the Student Club and Activities subsystem, outlining its purpose, core functionalities and role within the enterprise system. Related existing systems and architectural approaches used to support extracurricular activity management are also discussed.

2.3.1 Overview of the Student Club and Activities Subsystem

The Student Club and Activities Subsystem is designed to support and manage extracurricular engagement within UTHM. Extracurricular activities play a crucial role in student development by fostering leadership, teamwork and organizational skills. However, managing clubs, events, and related resources manually or through fragmented systems often results in problems such as inefficiencies, data inconsistency and administrative overhead.

This subsystem provides a digital platform to manage student clubs, memberships, events, attendance and club-owned or institution-provided assets. By integrating these functions into the enterprise system, the institution can improve operational efficiency, data accuracy and data transparency while enhancing the overall student experience.

The subsystem is intended to integrate with other enterprise components, such as the Staff Management System (SMS) for advisor data and the Facility Management System for venue bookings, ensuring seamless information flow across departments.

2.3.2 Subsystem Functional Scope

The Student Club and Activities Subsystem consists of several interrelated sub-modules that collectively support the lifecycle of club activities:

(A) **Club Profiles and Membership Management**

This module maintains club profiles, including club descriptions, objectives and advisor. It allows students to join or exit clubs digitally, while committee members are assigned specific roles such as President or Treasurer. Membership records are stored centrally to maintain governance, ease reporting and access control.

(B) **Event Proposal and Approval Workflow**

Club committees can submit event proposals through the system, specifying details such as event objectives, budget estimates and required resources.

These proposals and reports are routed digitally to the staff advisor in-command, who can approve or reject them. This workflow replaces paper-based approval processes and ensures traceability and accountability.

(C) **Event Registration and QR Code Generation**

The system generates a unique QR code for each event. This QR code is used for check-in, reducing manual attendance lists and minimizing human error. Students can register for club events.

(D) **Event Attendance Scanning and Tracking**

During events, students scan QR code to record attendance in real time. Attendance data is stored in the database and can be summarized through reports for club committees and administrative staff. This enables accurate tracking of student participation for co-curricular records.

(E) **Asset Borrowing System for Club Events**

Clubs may request to borrow institutional assets such as PA systems, chairs or equipment for events. This module supports asset requests, approval workflows, and return logging. It ensures that equipment usage is monitored and that assets are returned in a timely manner.

(F) **Club Finance and Budget Tracking**

This module records event budgets, club expenses, and fund usage using a simple ledger structure. It supports transparency in financial management and assists club treasurers and advisors in monitoring expenditures against allocated budgets.

2.3.3 Related and Previous Systems or Architectures

Traditionally, student club management in many institutions has relied on **manual or semi-digital systems**, such as paper application forms, google form based attendance, approvals sent through email and standalone web portals that display event listing and reminders. These approaches often lack integration, resulting in duplicated data, delayed approvals and limited reporting capabilities.

Some institutions have adopted **Student Information Systems (SIS)** or Learning Management Systems (LMS) with limited co-curricular modules, which may exist as standalone solutions or partially linked to other institution systems. While event announcements or attendance tracking can be supported, they typically do not provide comprehensive workflows for club governance, asset management or financial tracking.

From an architectural perspective, earlier systems often followed a **monolithic or silo-based architecture**, where club management functions operated independently from staff management, asset management or finance and e-payment systems. This limited scalability and set back enterprise-wide data consistency.

In contrast, modern enterprise systems adopt **modular and service-oriented architectures**, enabling subsystems like Student Club and Activities to integrate seamlessly with other institutional systems. By leveraging centralized databases, role-based access control and workflow automation, the proposed subsystem aligns with contemporary enterprise architecture principles and supports long-term scalability and maintainability.

2.4 Technology Used

This section describes the technologies used in the development of Student Clubs and Activities Management System and explains why they are suitable for the project.

2.4.1 SAP Build Apps

SAP Build Apps is a **low-code or no-code application development platform** provided by SAP. It allows developers to create enterprise applications using visual tools instead of complex programming. Users can design user interfaces by dragging and dropping components and define application logic using visual workflows.
SAP Build Apps is suitable for academic and university projects because it:

- Provides **fast prototyping**, allowing students to develop applications in a short time.
- Reduces the need for advanced programming skills.
- Supports **visual UI design**, which improves usability and user experience.
- Allows easy connection to databases and backend services.

In this project, SAP Build Apps is used to develop Student Clubs and Activities Management System prototypes, including club registration, event management, attendance tracking and asset borrowing modules. The platform supports UTHM’s vision of a modern, integrated campus system and demonstrates how enterprise applications can be developed efficiently.

2.4.2 Database Technology

SAP Build Apps uses a cloud-based database provided by SAP Business Technology Platform (BTP) to store application data. The database supports structured data storage, security and future integration with other enterprise systems. For the Student Clubs and Activities Management System, the database stores important information such as:

- Club profiles and descriptions
- Student membership records and roles
- Event details and attendance records
- Asset borrowing and return transactions

Using a structured database ensures:

- Data consistency, where information is stored in one central location
- Reduced data duplication across different systems
- Easy management of relationships between data
- Better support for reporting and future system enhancements

The database is designed to logically integrate with UTHM’s existing enterprise systems, such as Student Information System (SMP) and Staff Information System (SMS).

2.4.3 Supporting Frameworks and Technologies

Several supporting technologies are applied to enhance the functionality and efficiency of the system.

QR Code Technology

QR codes are used for **event attendance tracking**. In SAP Build Apps, QR code technology works by generating a unique QR code for each event registration, which is scanned during the event to automatically verify and record student attendance in the database. This method improves attendance accuracy, reduces manual errors and saves administrative time.

Role-Based Access Control (RBAC)

Role-based access control ensures that users can only access features related to their roles. For example:

- Students can join clubs and register for events
- Club committee members can manage events and assets
- Advisors can approve event proposals
- Administrators can monitor overall system activities

This improves system security and prevents unauthorized access.

Web and Mobile Interfaces

The system is designed to be accessible through web and mobile interfaces, allowing students and staff to use it anytime and anywhere. This improves convenience, increases participation in activities and supports UTHM's smart campus environment.

The selected technologies are suitable for this project because they:

- Support enterprise architecture principles
- Improve system efficiency and data management
- Easy to use and maintain
- Align with UTHM's digital ecosystem and future integration plans

Overall, these technologies provide a strong foundation for developing a reliable and scalable Student Clubs and Activities Management System.

2.5 Summary

This chapter reviewed literature related to the Enterprise Architecture, university enterprise systems and student activity management subsystems. Previous studies show that applying EA principles in higher education improves system integration, reduces data silos and improves service delivery. Research on student clubs and activities systems highlights the need for centralized, digital and integrated solutions. The reviewed technologies, specificallySAP Build Apps, centralized databases and QR-based attendance management, provide strong justification for the design of the proposed system. The literature supports the project's objectives to develop an EA-aligned subsystem that enhances student engagement while integrating logically with UTHM's existing digital ecosystem.

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in the development of the UTHM System, namely the Waterfall model. The selected methodology and its implementation are discussed in detail, highlighting how each phase is applied throughout the system development process. In addition, a project planning schedule is provided to illustrate the overall timeline and major milestones of the project.

3.2 Chosen Methodology

The system development approach used in this project is the Waterfall Methodology. The Waterfall model is an organized and sequential technique in which development tasks are divided into clearly defined phases with each phase completed before proceeding to the next.

The Waterfall methodology was chosen since the Students Clubs and Activities Management System's requirements were identified early on. The project's objectives are to analyze UTHM’s current digital systems, apply Enterprise Architecture (EA) concepts and create a prototype with SAP Build Apps. The Waterfall approach successfully supports the systematic planning, precise documentation and organized workflow needed for these tasks.

Furthermore, the Waterfall methodology encourages detailed documentation throughout the development process, including requirements planning, system design and implementation planning. This is useful for the project clarity, traceability as well as evaluation of project outcomes. Managing the project timeline and guaranteeing that all deliverables are finished on time further benefits the well-defined phase-by-phase structure.

3.3 Phases of Methodology

The Waterfall approach used in this project includes various steps such as the requirements gathering, system design, implementation, testing and deployment. Every stage is essential to the development of the proposed system.

3.3.1 Requirements Gathering Phase

The main goal of this first phase is understanding the current digital environment at UTHM and determining system requirements. Information about existing systems including the Student Information System (SMP), Staff Information System (SMS) and MyUTHM mobile application is gathered during this phase through observation, document analysis and conversations.

The purpose of this phase is to define both functional and non-functional requirements for the proposed Student Clubs and Activities Management System. A list of necessary system features, stakeholder identification and an understanding of current business procedures and system constraints are examples of key outcomes. This stage guarantees that the suggested system complies with UTHM's digital systems and meets actual operational requirements.

3.3.2 System Design Phase

In the system design phase, the overall structure of the proposed system is created based on Enterprise Architecture (EA) principles. This includes creating system workflows, database structure, module interactions and basic user interface layouts in addition to defining the business, application, data and technology layers. This phase also addresses integration issues with current UTHM systems and the result is a comprehensive system design blueprint that may be used as a reference for system implementation.

3.3.3 Implementation Phase

Using SAP Build Apps, the system prototype is developed during the implementation phase based on the required design. During this phase, the development environment is established and system modules such as club registration, event management, attendance tracking and asset borrowing are developed and implemented. SAP Build Apps' visual development tools and workflow logic are utilized to create a functional subsystem within the project deadline.

3.3.4 Testing Phase

The testing process ensures that the developed system meets the given requirements and operates properly. Verifying system functionality and making sure all modules function as intended are examples of testing procedures. User-based testing will also be carried out to assess usability and workflow effectiveness and discover flaws or errors that are resolved before the system is deployed.

3.3.5 Deployment Phase

In the deployment phase, the finished system prototype is ready for testing and assessment. The system is set up in a controlled setting and made available to the intended users including administrators, club committee members and students. This stage completes the development process and demonstrates how the suggested enterprise solution may be incorporated into UTHM's digital systems.

3.3.6 Maintenance Phase

The maintenance phase focuses on assuring the system's continuing reliability and performance after deployment. This includes monitoring system operation, resolving bugs, updating system functionality and making minor changes as needed. This phase emphasizes the necessity of system sustainability and flexibility for future enhancements and integrations into UTHM's organizational environment.

3.4 Project Planning Schedule

Figure 1 presents the project planning schedule based on the Waterfall model.

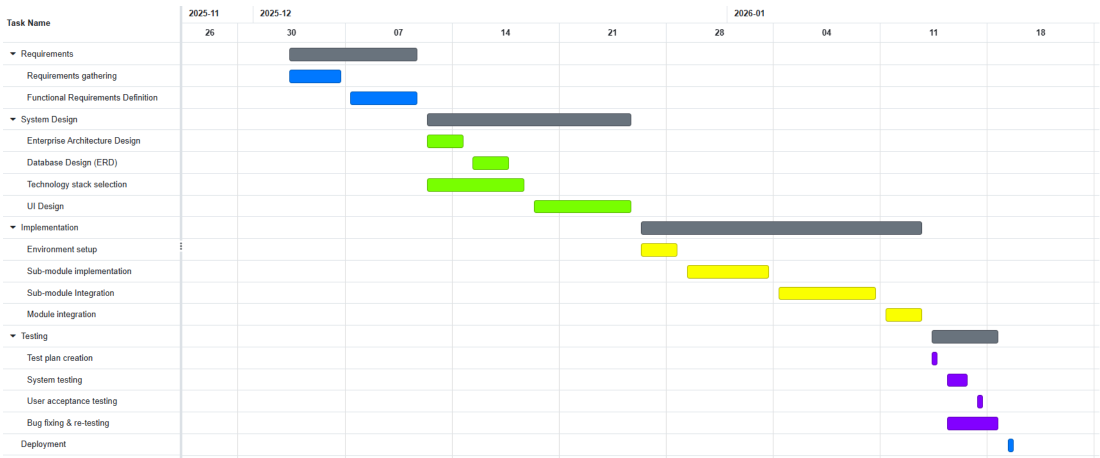

Figure 1 Gantt Chart

The project began with the requirements gathering phase. On 3rd December 2025, we visited UTHM to understand the requirements and gather relevant information. Based on the findings, the functional requirements, modules, and sub-modules were identified. Approximately one week was allocated for this phase.

The next phase is system design, during which the technical aspects of the system are considered. A total of two weeks is allocated for this phase. Activities include system architecture design, database design, and technology stack selection. In addition, a general user interface (UI) is designed to serve as a reference for further development. By the end of this phase, there is a clear understanding of how the system will function and appear, and is ready to proceed to implementation.

Following the design phase, the system is implemented using the selected technology stack. Initially, the development environment is set up. Sub-modules are then developed and integrated, followed by the integration of all modules to form a complete enterprise system. This phase is allocated approximately three weeks.

After implementation, the system undergoes testing prior to deployment and use by actual users. Both developers and users participate in the testing process to ensure the system is functional and meets stakeholder expectations. Throughout the testing, any identified bugs or errors are fixed.

Finally, once all requirements are satisfied and the system is stable, it is deployed and released.

3.5 Summary

The Waterfall model is used to guide the system development process. It consists of the requirements, design, implementation, testing, deployment and maintenance phases. A gantt chart is used as reference to plan the project timeline and to monitor and trace the progress of the development.

CHAPTER 4
ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the system analysis and design for the proposed enterprise subsystem at Universiti Tun Hussein Onn Malaysia (UTHM). It aims to analyse existing business processes and digital systems related to student clubs and activities within UTHM, in order to identify inefficiencies, data fragmentation and integration issues. The chapter begins by outlining UTHM's organizational structure to provide institutional context, followed by an analysis of the current system and its limitations. A comparison between the existing and proposed systems is then presented to highlight improvements in efficiency, accuracy, security and user experience. System requirements are gathered using interviews, questionnaires and observation and are formalized into functional and non-functional requirements. Finally, system design artefacts including use case diagrams, user stories, sequence diagrams and activity diagrams are presented to model system behaviour and interactions. This chapter serves as a crucial link between the conceptual foundation and the implementation phase by translating enterprise needs into structured design models aligned with Enterprise Architecture principles.

4.2 Company Organization Structure

4.2.1 Background of Universiti Tun Hussein Onn Malaysia (UTHM)

Universiti Tun Hussein Onn Malaysia (UTHM) is a public technical university located in Batu Pahat, Johor. UTHM focuses on engineering, technology and vocational education and serves approximately 18 000 students. The university operates through multiple academic and administrative departments that work together to support teaching, learning, research and student development.

To manage its operations efficiently, UTHM adopts a structured organizational hierarchy that ensures clear authority, responsibility and coordination among departments. This structure plays an important role in the successful implementation and operation of digital systems such as the Student Clubs and Activities Management System.

4.2.2 UTHM Organization Chart

Figure 2 shows the department involved in managing the Student Clubs and Activities Management System.

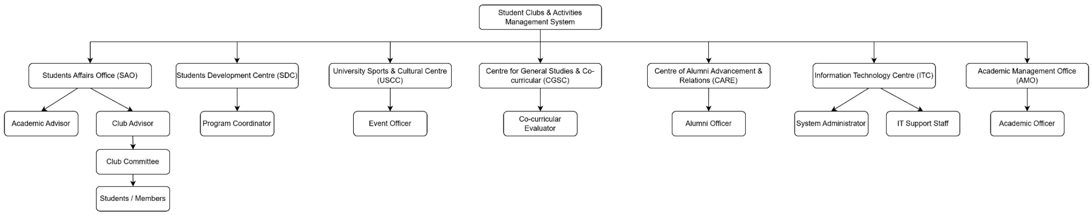

Figure 2 UTHM Organization Chart

4.2.3 Key Departments Related to the System

Table 1 shows key UTHM departments, their roles and the system modules they use in managing student clubs and activities.

Table 1 UTHM Department, Roles and System Modules

UTHM Department	Sub-Module	Role
Students Affairs Office (HEP)	• Club Profiles & Membership Management • Event Proposal & Approval Workflow • Club Finance & Budget Tracking	Manage club registration, member roles, approves events, monitors club finance and budget compliance
Students Development Centre	• Event Proposal & Approval Workflow • Event Registration + QR Code Generation	Evaluates student involvement, leadership roles; uses registration data for development programs
University Sports and Cultural Centre	• Event Registration + QR Code Generation • Event Attendance Scanning & Tracking	Coordinates sports or cultural events; tracks participation and attendance
Centre for General Studies and Co-curricular	• Event Attendance Scanning & Tracking	Retrieves attendance for co-curricular credit evaluation
Centre of Alumni Advancement and Relations (CARE)	• Event Registration + QR Code Generation • Event Attendance Scanning & Tracking	Coordinates joint student-alumni events; track participation
Information	All modules	Provides technical

Technology Centre		support, ensures system functionality, QR code generation, data storage and security
Academic Management Office	● Event Attendance Scanning & Tracking	Ensures participation data aligns with academic records for co-curricular recognition

4.3 Current System Analysis

Currently, UTHM operates several computerized systems to support its academic and administrative operations. These systems are used by different stakeholders such as students, academic staff and administrative personnel for daily tasks.

Firstly, the Student Information System (SMP) is used to manage student-related data, like their biodata and examination results. However, SMP primarily focuses on academic administration and has limited integration within non-academic systems such as hostel management and student club activities. This limitation makes it difficult for the administrators to overview the overall performance of the students because it can mainly support academic-related monitoring only.

Secondly, the Staff Information System (SMS) is used to manage staff-related information, including biodata, leave records and employment documentation. This system supports basic human resource functions that are useful for the university administration to manage the staff records. However, the system has limited integration with other administrative or academic platforms, resulting in repeated data entry and fragmented staff information.

Thirdly, the UTHM Single Sign-On (SSO) portal provides a centralized authentication gateway that allows students and staff to access across multiple systems using specific user credentials. Although SSO improves user convenience by connecting all the services together in one portal, it does not facilitate backend data integration between systems. Each application maintains separate databases and workflows, causing limited data sharing.

Finally, the MyUTHM mobile application offers selected digital services for both students and staff. For instance, users are able to access basic information as in SMP and SMS, while students can view attendance records and track the shuttle bus on campus. However, many student-facing processes, including hostel applications, student club and academic advising are not available within the application. As a result, users need to switch to different web portals to access all these services, reducing the overall effectiveness of the mobile application.

4.3.1 Key Issues in the Current System

Based on the analysis of UTHM's existing digital systems, several key issues have been identified:

1. System Fragmentation and Data Silos
   Multiple systems operate independently with limited integration, resulting in inconsistent data updates and duplicate information across various platforms.
2. Manual and Semi-Automated Processes
   Certain processes, such as academic advising and student club management still rely on manual processes, indirectly increasing the processing time to the request from users.
3. Limited Mobile Integration
   MyUTHM didn't cover all the services provided to the students and require them to access different portals.

4.4 Comparison between Existing and Proposed System

A comparison of the proposed system and the current UTHM system is shown in Table 2. Relevant aspects like speed, accuracy, cost, security and user experience are the main focus of the comparison.

Table 2 Comparison of The Proposed System and The Current UTHM System

Aspect	Existing System	Proposed System
Speed	Delays occur because different procedures such as event approval, attendance tracking and asset booking are handled using separated or semi-automated tools.	End-to-end procedures are fully automated, enabling speedier processing and real-time updates.
Accuracy	Data is stored across multiple systems, increasing the risk of data inconsistency and duplication.	A centralized database with automated validation assures high data accuracy and consistency.
Cost	Higher operational cost due to manual documentation, repeated data entry and administrative workload.	Reduced operational cost through paperless processes, system automation and efficient resource management

Security	Access control and data management differ across the platform, making it challenging to implement consistent security standards.	Role-based access control and centralized data storage enhance the data protection, access control and system security.
User Experience	Users need to switch between various platforms to manage clubs, events and attendance, which results in fragmented workflows and lower usability.	Integrated web and mobile interfaces provide user-friendly experience for students, club committees and administrators.

The comparison shows that while the current existing system at UTHM provides some digital features, such functionalities are being deployed separately across many platforms. The new system, Student Clubs and Activities Management System, improves efficiency and accuracy through a smooth integration of related activities into a central enterprise system. The new system will result in lower cost of operation, greater data security through role-based security and increased user accessibility via a centralized system that fits the enterprise architecture.

4.5 System Requirements

This section lists the functional requirements and the non functional requirements of the Student Club and Activities Management Subsystem.

4.5.1 Functional Requirement

The functional requirements are broken down into subsystems for readability.

General Functional Requirements

These are the general functional requirements that should be implemented and kept consistent across the system.

Table 3.1 General Functional Requirements

FR1.1	The system shall **validate the user input** including but not limited to email address format, identity card number format, matric number format, staff number format, phone number format and uploaded file format.
FR1.2	The system shall display **error messages for invalid input** including but not limited to missing required fields, incorrect input format, invalid QR code.

FR1.3	The system shall display a **confirmation message** to the user after successfully updating the database. Including but not limited to: ● Successful submission of membership application, club application, event proposal ● Successful records the approval or rejection of membership application, club application, event proposal ● Successful updates the club information ● Successful registration of events.
DR1.4	The system shall display an **error message** that is understandable by non technical users for unsuccessful operations.
FR1.5	The system shall **notify related users via email** upon successful submission of application and approval or rejection of application. ● Notify Club Committee when a user submits membership request ● Notify applicant of the result of membership application / club application ● Notify Club advisors when club committee of the supervised club submitted an event proposal
FR1.6	The system shall ensure that all **data is stored securely and comply with relevant data protection regulations**.
FR1.7	The system shall ensure that **all buttons are functionable**.
FR1.8	The system shall ensure 3that **all forms are functionable**.
FR1.9	The system shall record all actions with **timestamps**.

Submodule 1 Club Profiles and Membership Management

The following is the functional requirements for the submodule club profiles and membership management.

Table 3.2 Functional Requirements for Club Profiles and Membership Management

FR2.1	Users should be able to **browse through the list of registered clubs** in UTHM, along with its information such as club name, description, category and status, and should be able to filter the clubs by category and search for a club.
FR2.2	Users should be able to **apply for membership** of a club.

FR2.3	Users, specifically club committee, club advisor and applicant, should be **notified** when the applicant successfully submits the membership application.
FR2.4	Users, specifically club committee and club advisor, should be able to **browse through the list of membership applications**, including the applicant's name, email, contact number and application description.
FR2.5	Users, specifically club committee and club advisor, should be able to **approve and reject the member application**.
FR2.6	Users, specifically club committee and club advisor, should be able to **update the club information**.
FR2.7	Users should be able to **register a new club** by submitting a form.
FR2.8	Users, specifically the applicant and associated club advisor, should be **notified** when the application successfully submits the club registration.
FR2.9	Users, specifically the club advisor, should be able to **view the club registration application** and **approve or reject the club registration**.
FR2.10	Users, specifically the HEP officer, should be able to **view the club registration application** and **approve or reject the club registration** that is approved by the associated club advisor.
FR2.11	Users, specifically the club advisor, should be able to **monitor club activities** under his or her supervision, including the members' roles and activities logs
FR2.12	Users, specifically the HEP officer, should be able to **monitor registered club activities**, including the members' roles and activities logs.

Submodule 2 Event Proposal and Approval Workflow

The following is the functional requirements for the submodule Event Proposal and Approval Workflow.

Table 3.3 Functional Requirements for Event Proposal and Approval Workflow

FR3.1	Users, specifically the club committee, should be able to **submit event proposals** through the system.
FR3.2	Users, specifically the Club Advisors, should be able to **view the event proposals** and **approve or reject the event proposal** that is submitted by the club committee.

FR3.3	Users, specifically the Club Advisors, should be able to **write comments and remarks** about the event proposal submitted by the club committee.
FR3.4	Users, specifically club committee, should be able to **view the status (Pending, Approved and Rejected) of the submitted proposal**, including advisor remarks.

Submodule 3: Event Registration and QR Code Generation

The following is the functional requirements for the submodule Event Registration and QR Code Generation

Table 3.4 Functional Requirements for Event Registration and QR Code Generation

FR4.1	Users should be able to **view the list of events** that is approved, along with its information such as event name, description, date, time, venue, organiser and fee, and should be able to filter the events by category, organiser, date and eligibility.
FR4.2	Users should be able to **register for an event**.
FR4.3	The system shall display a **warning message** if the user registered in an event that has clashing time with other events registered by the user.
FR4.4	The system shall only allow the user to **register one event once** only to prevent duplicated registration.
FR4.4	The system shall be able to **generate a QR code** that encodes the information of the registration if the event is a paid event.
FR4.5	Users should be able to **review the list of events registered**.
FR4.6	Users should be able to **cancel the registration** of a registered event.
FR4.7	The systems should be able to **determine if the user has the authority to view further details** of the registration.
FR4.8	Users, specifically the club committees, club advisors and HEPA officers, should be able to **view the participants registered** for the event.

Submodule 4: Event Attendance Scanning and Tracking

The following is the functional requirements for the submodule event attendance scanning and tracking.

Table 3.5 Functional Requirements for Event Attendance Scanning and Tracking

FR5.1	Users should be able to **scan attendance QR code** to record the attendance.
FR5.2	The system should be able to **activate the device camera** to scan QR codes.
FR5.3	The system should be able to **decode QR codes and retrieve relevant information** from the database.
FR5.4	Users, specifically the club committee and club advisors should be able to **view the attendance of events** organised by the associated club, while the HEP officers should be able to view the attendance of all events.

Submodule 5: Asset Borrowing System for Club Events

The following is the functional requirements for the submodule Asset Borrowing System for Club Events.

Table 3.6 Functional Requirements for Asset Borrowing System for Club Events

FR6.1	Users should be able to **view a list of assets** including their name, storage venue, status, rental and deposit, and should be able to filter the assets by category, status and rental.
FR6.2	Users, specifically students, should be able to **submit rental requests** by providing information including but not limited to the event name, date, duration and purpose of rental.
FR6.3	The system must be able to automatically **check for schedule conflict and prevent double-booking** of any assets for the same time slot.

Submodule 6: Club Finance and Budget Tracking

The following is the functional requirements for the submodule Club Finance and Budget Tracking.

Table 3.7 Functional Requirements for Club Finance and Budget Tracking

FR7.1	Users, specifically club committee, should be able to **add and submit a financial record**.
FR7.2	Users, specifically club advisors, should be able to **view the financial record** of the clubs under their supervision and add **comments and notes about the record**.

FR7.3	Users, specifically the HEP officers, should be able to view the club financial reports.

4.5.2 Non Functional Requirement

The following are the non-functional requirements for the Student Club and Activities Management Subsystem.

Table 3.8 Non Functional Requirements

NFR1	The system shall support up to 20,000 concurrent users without the degradation of performance.
NFR2	The system shall have an uptime of 99.99% manually.
NFR3	The system shall be available 24x7 excluding the maintenance window.
NFR4	The system shall return results within 2 seconds for any search and filters.
NFR5	The system shall not exceed 75% CPU utilization under peak load.
NFR5	The application shall be able to execute on various browsers including but not limited to FireFox, Bing, Opera, Chrome and Explorer.
NFR6	The system must allow deployment of bug fixes within 12 hours of identifying critical issues.
NFR7	The data stored must aligned with the Personal Data Protection Act 2010 (PDPA)
NFR8	The system must include role-based access control (RBAC) to secure user permissions.
NFR9	Error messages must be easily understandable by non-technical users.
NFR10	Response times for critical functions must not exceed 3 seconds during peak usage.
NFR11	Disaster recovery must include the ability to restore operations within 30 minutes in case of hardware failure.
NFR12	The system must include intrusion detection/prevention systems (IDPS) to monitor for malicious activities.
NFR13	Backup power systems (e.g., UPS, generators) must be available to support at least 8 hours of operation.

NFR14	The system’s user interface must be intuitive and provide training for non-technical users.
NFR15	Onboarding and training resources must be available for new users and administrators.

4.6 System Design

This section presents the system design of the proposed subsystem, focusing on how the functional requirements are translated into system interactions and workflows.

4.6.1 Use Case Diagram

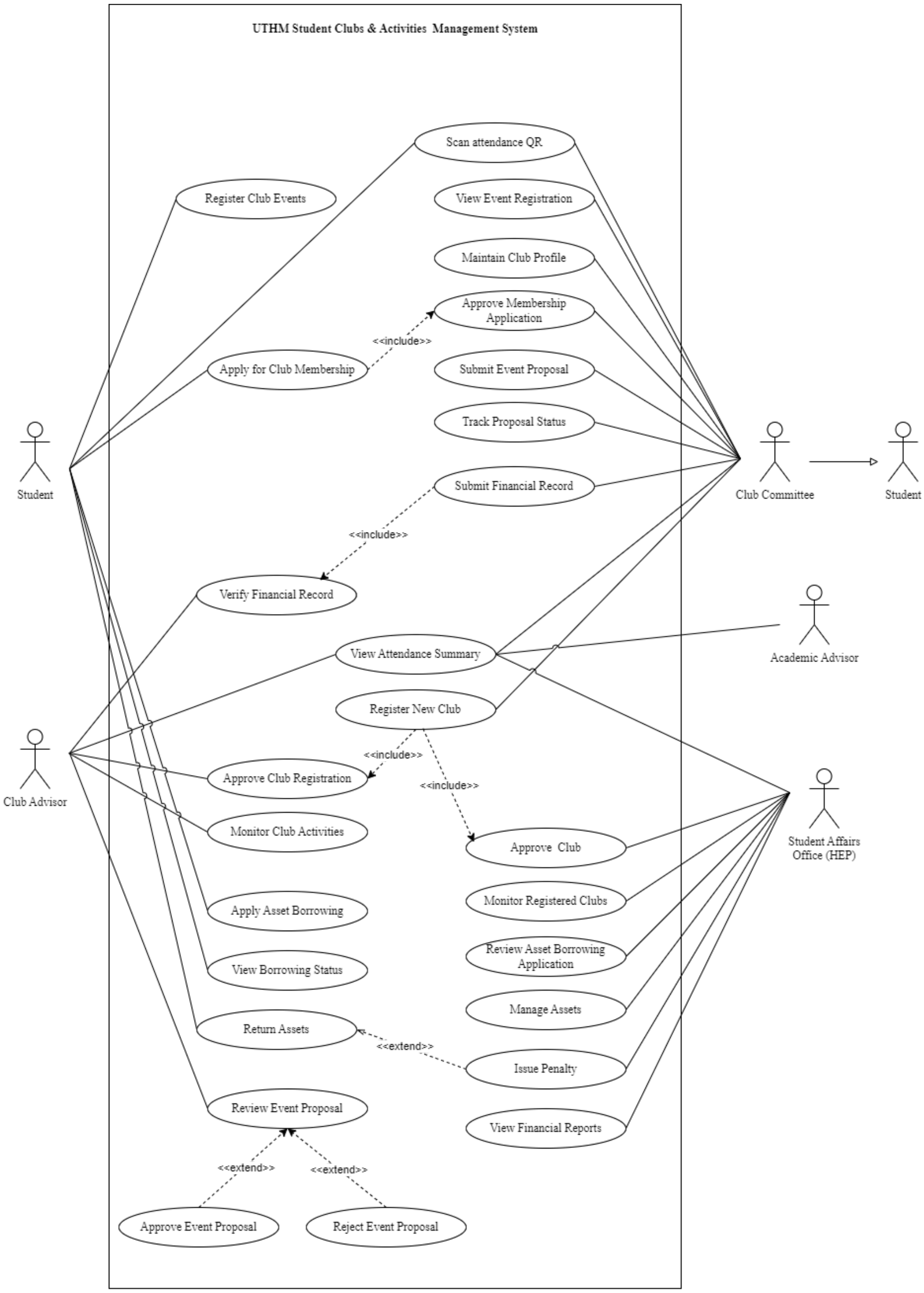

Figure 3 Use Case for UTHM Student Clubs and Activities Management System

4.6.2 User Story

This section explains the flow of each user story, supported by the corresponding activity and sequence diagrams.

4.6.2.1 US01 Apply for Club Membership

Table 4.1 shows the user story description, followed by the sequence diagram in Figure 3.1.1 and the activity diagram in Figure 3.1.2.

Table 4.1 User Story Description for Apply for Club Membership

User story: Apply for Club Membership
ID: US 01
User Story Description: **As a** Student **I want** to apply for membership in a selected club **So that** I can become a member of the club
Flow of events: 1. Student selects “View Clubs”. 2. System retrieves and displays club profiles (name, description, category, status). 3. Student selects a club. 4. Student submits membership application. 5. System records request. 6. System notifies Club Committee.
Alternative flow: 1. If a student cancels the application, there is no request stored.
Acceptance Criteria: 1. Application is successfully submitted. 2. Club Committee receives notification. Precondition: Student has selected a club. Postcondition: Membership request is pending approval.
Exception flow: If a student has already applied, the system will display a warning.

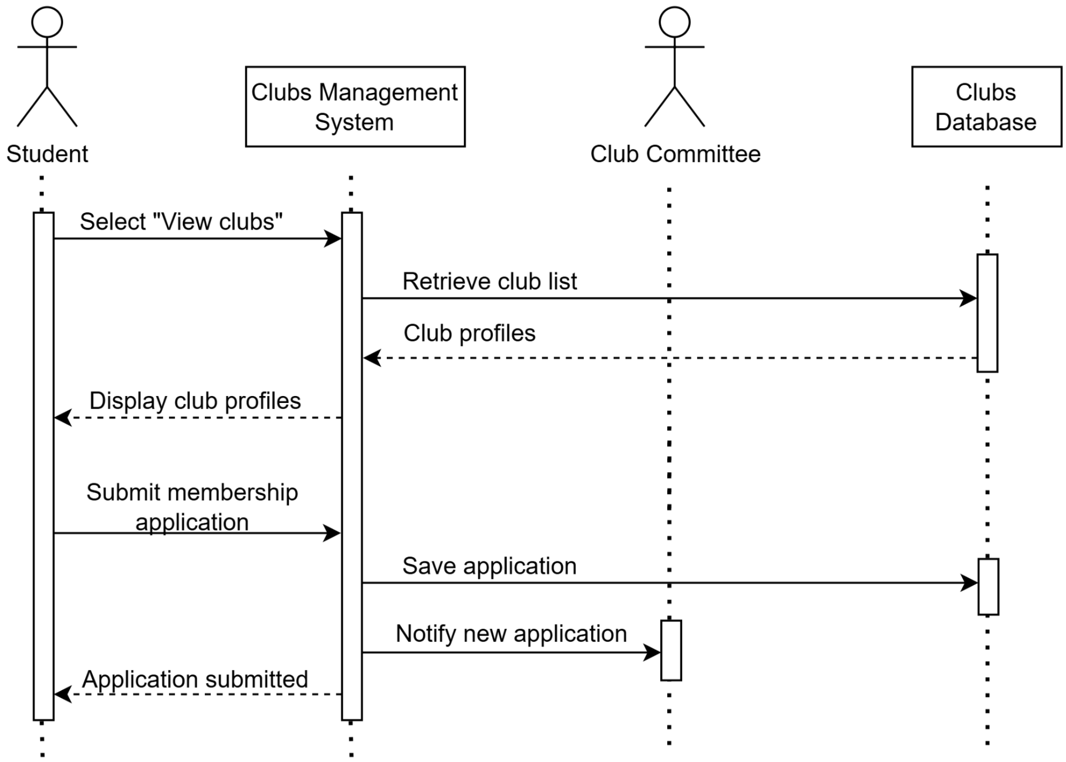

Figure 3.1.1 Sequence Diagram for Apply for Club Membership

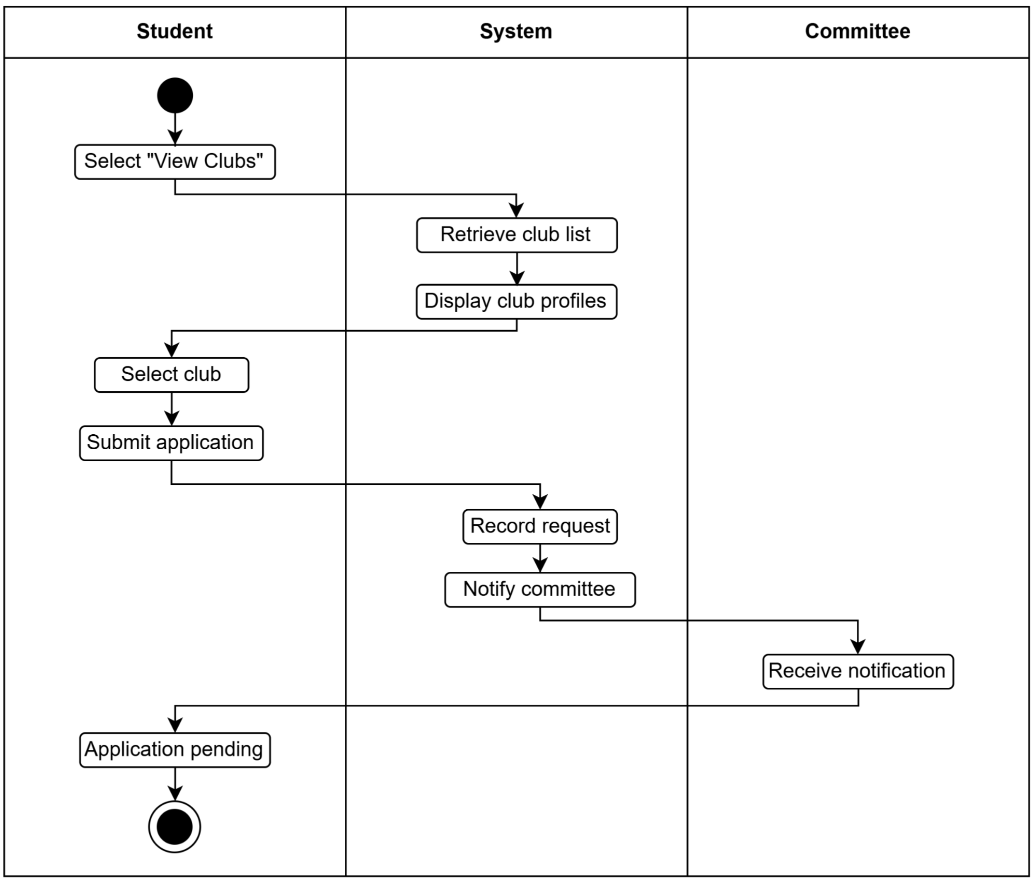

Figure 3.1.2 Activity Diagram for Apply for Club Membership

4.6.2.2 US02 Approve Membership Application

Table 4.2 shows the user story description, followed by the sequence diagram in Figure 3.2.1 and the activity diagram in Figure 3.2.2.

Table 4.2 User Story Description for Approve Membership Application

User story: Approve Membership Application
ID: US 02
User Story Description: **As a** Club Committee Member **I want** to approve or reject membership requests **So that** only eligible students become club members.
Flow of events: 1. Committee views pending applications. 2. Committee reviews application. 3. Committee approves or rejects. 4. System updates status and notifies student.
Alternative flow: 1. Committee requests additional information.
Acceptance Criteria: 1. Membership status is updated. 2. Student receives notification. Precondition: Student has submitted application. Postcondition: Student becomes member or is rejected.
Exception flow: If the system fails, the decision will not be saved.

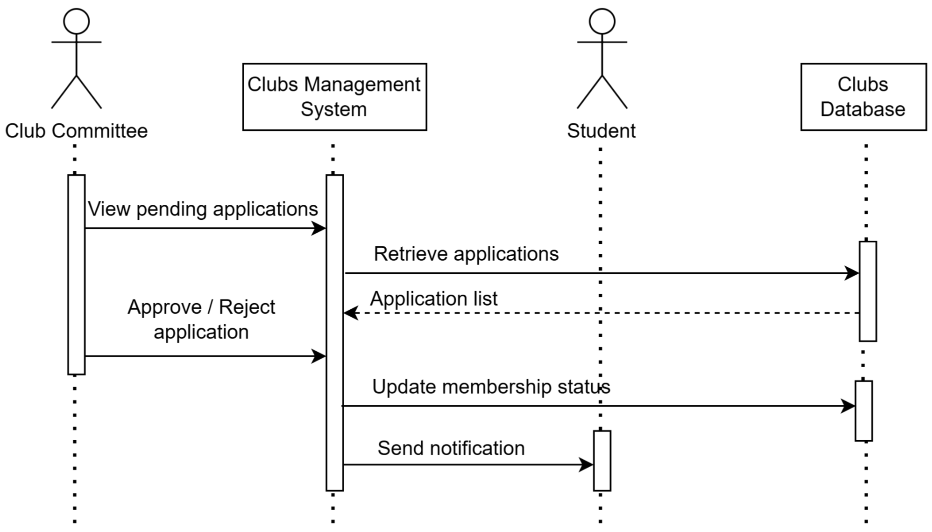

Figure 3.2.1 Sequence Diagram for Approve Membership Application

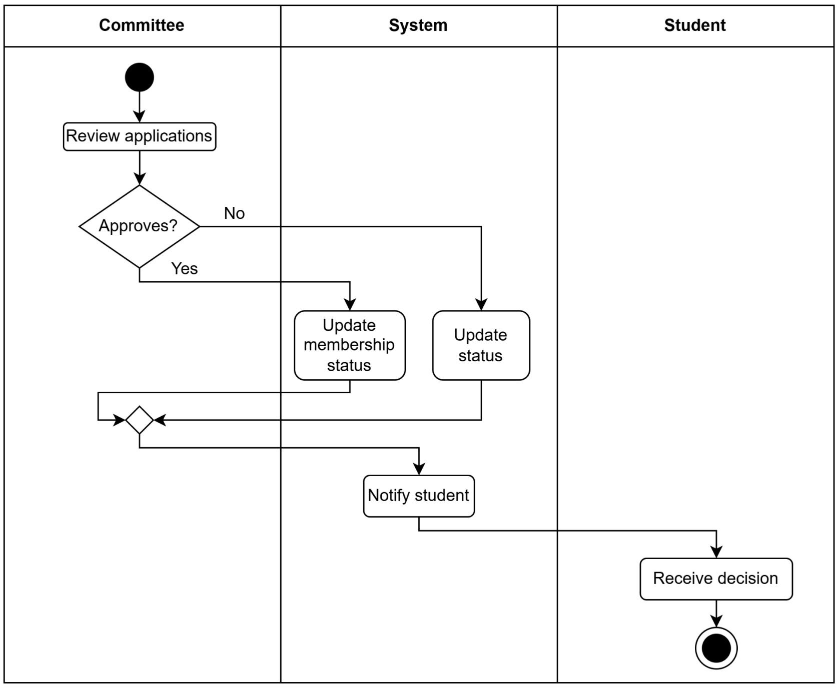

Figure 3.2.2 Activity Diagram for Approve Membership Application

4.6.2.3 US03 Maintain Club Profile

Table 4.3 shows the user story description, followed by the sequence diagram in Figure 3.3.1 and the activity diagram in Figure 3.3.2.

Table 4.3 User Story Description for Maintain Club Profile

User story: Maintain Club Profile
ID: US 03
User Story Description: **As a** Club Committee Member **I want** to manage club information and member records **So that** club details remain accurate and updated.
Flow of events: 1. Committee selects club profile. 2. Updates club info. 3. Views member list. 4. Assigns or updates roles. 5. Adds or removes members. 6. Saves changes.
Alternative flow: 1. Committee discards changes.
Acceptance Criteria: Precondition: Club is registered and active. Postcondition: Updated club and membership data saved.

Exception flow: If the data is invalid, the system rejects the update.

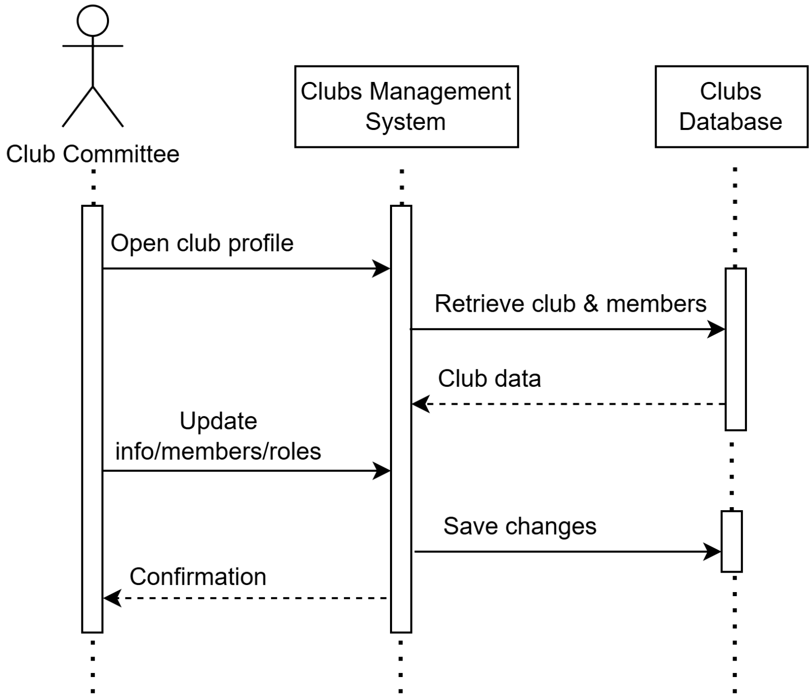

Figure 3.3.1 Sequence Diagram for Maintain Club Profile

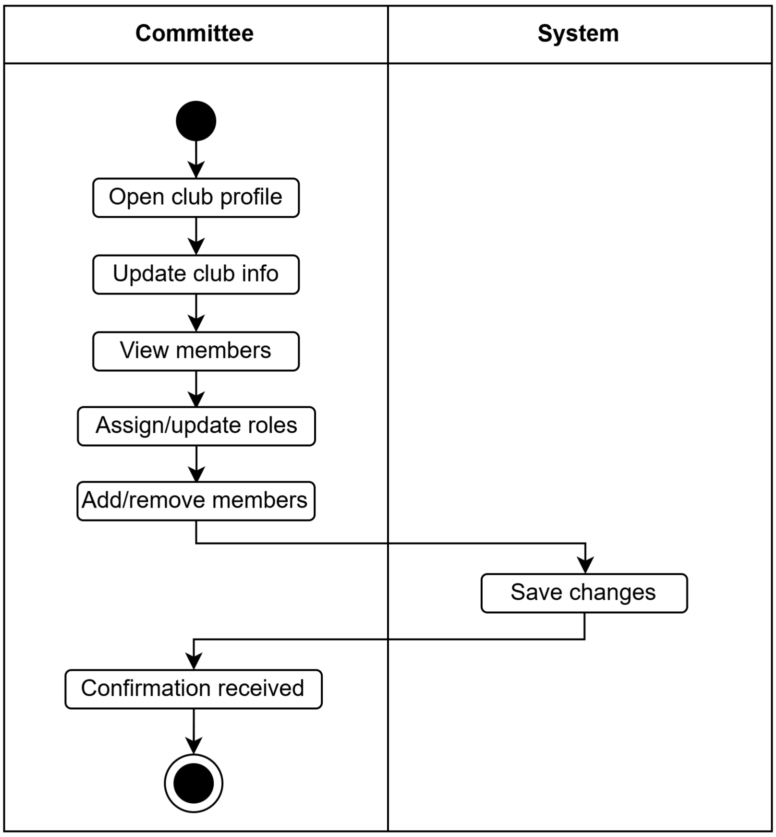

Figure 3.3.2 Activity Diagram for Maintain Club Profile

4.6.2.4 US04 Register New Club

Table 4.4 shows the user story description, followed by the sequence diagram in Figure 3.4.1 and the activity diagram in Figure 3.4.2.

Table 4.4 User Story Description for Register New Club

User story: Register New Club
ID: US 04
User Story Description: **As a** Club Committee Member **I want** to submit a new club registration **So that** the club can be officially recognized.
Flow of events: 1. Committee fills registration form. 2. Submits application. 3. System forwards to Club Advisor.
Alternative flow: 1. If the form is incomplete, the system requests correction.
Acceptance Criteria: 1. Registration request is successfully submitted. Precondition: Committee completed form. Postcondition: Registration enters advisor review.
Exception flow: If the system errors, the submission fails.

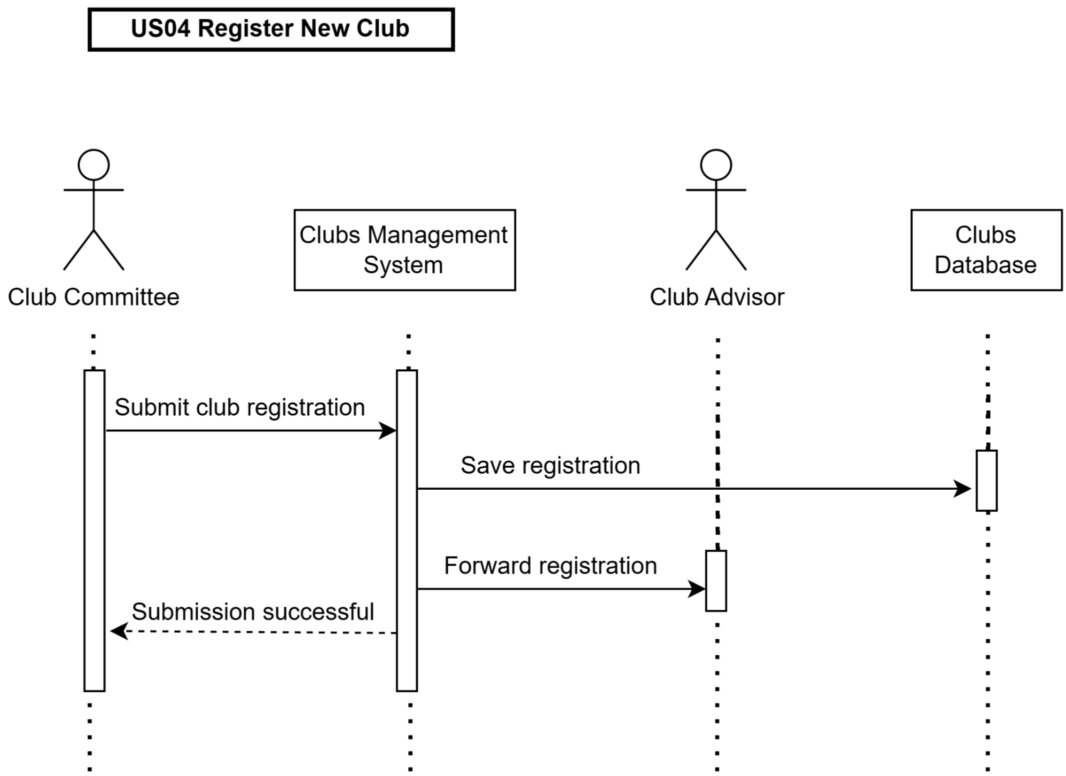

Figure 3.4.1 Sequence Diagram for Register New Club

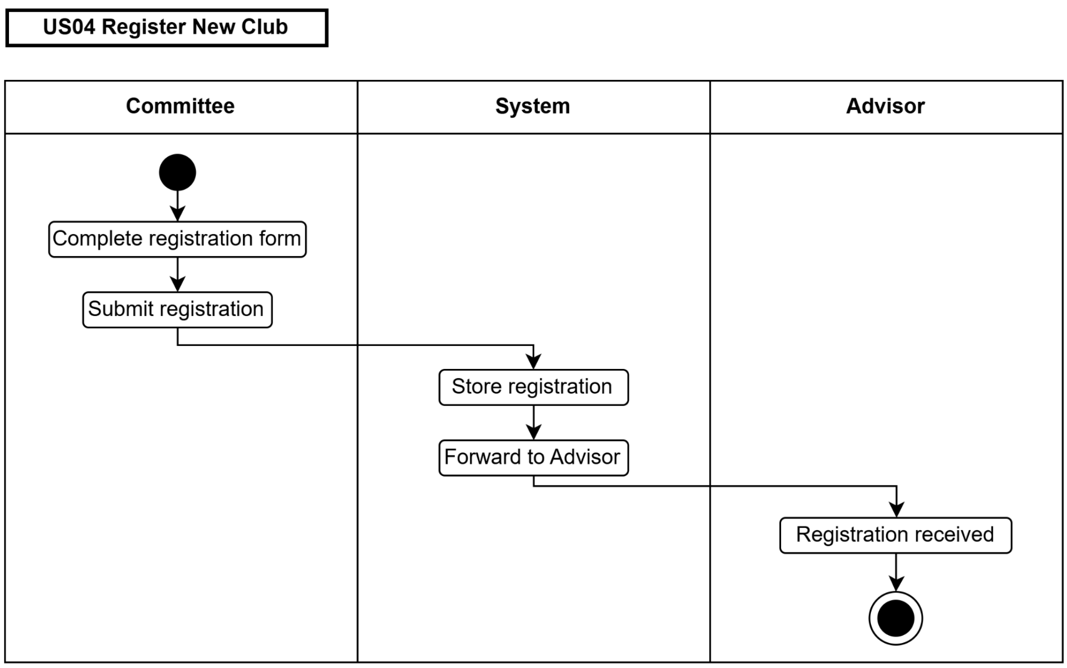

Figure 3.4.2 Activity Diagram for Register New Club

4.6.2.5 US05 Approve Club Registration

Table 4.5 shows the user story description, followed by the sequence diagram in Figure 3.5.1 and the activity diagram in Figure 3.5.2.

Table 4.5 User Story Description for Approve Club Registration

User story: Approve Club Registration
ID: US 05
User Story Description: **As a** Club Advisor **I want** to review and validate new club proposals **So that** only suitable clubs proceed for approval.
Flow of events: 1. Advisor reviews proposal. 2. Approves and forwards to HEP.
Alternative flow: 1. Advisor rejects and returns proposal.
Acceptance Criteria: Precondition: Registration submitted by committee. Postcondition: Registration is forwarded to HEP if approved.
Exception flow: If there are missing documents, the advisor cannot proceed.

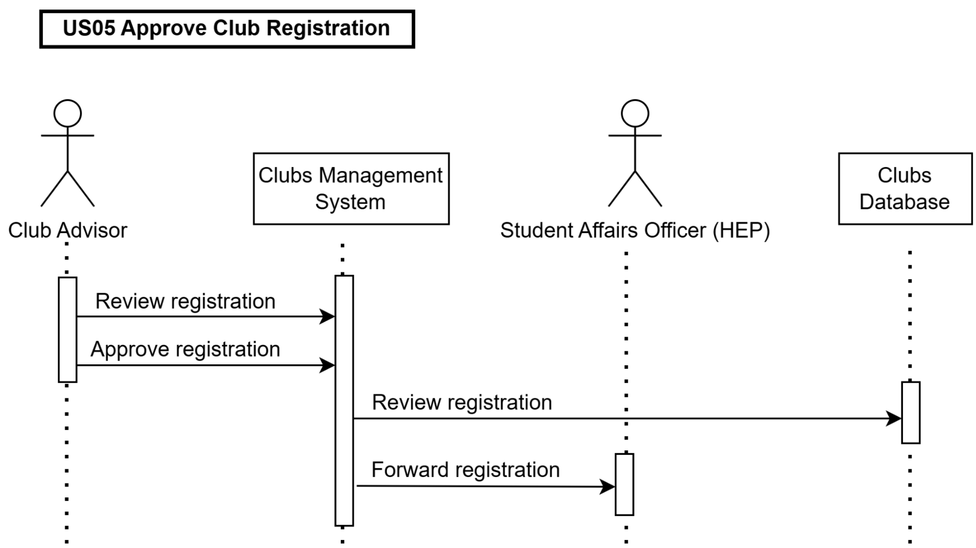

Figure 3.5.1 Sequence Diagram for Approve Club Registration

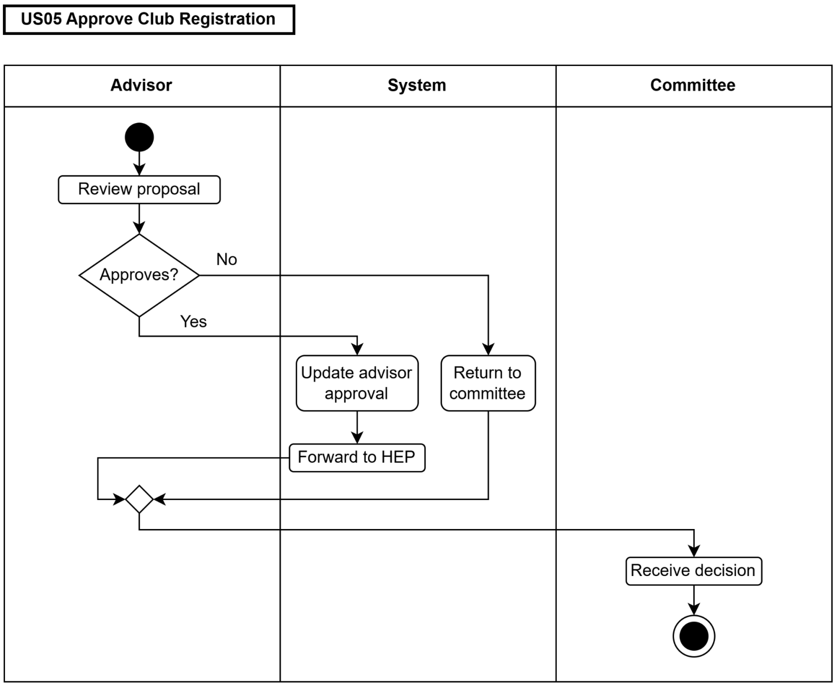

Figure 3.5.2 Activity Diagram for Approve Club Registration

4.6.2.6 US06 Monitor Club Activities

Table 4.6 shows the user story description, followed by the sequence diagram in Figure 3.6.1 and the activity diagram in Figure 3.6.2.

Table 4.6 User Story Description for Monitor Club Activities

User story: Monitor Club Activities
ID: US 06
User Story Description: **As a** Club Advisor **I want** to monitor club activities **So that** club operations comply with university policies.
Flow of events: 1. Advisor accesses dashboard. 2. Reviews roles and activity logs. 3. Provides feedback or flags issues.
Alternative flow: 1. If there are no activities, the dashboard shows none.
Acceptance Criteria: 1. Advisor can view activities and roles. Precondition: Club is active. Postcondition: Club is monitored and guided.
Exception flow: -

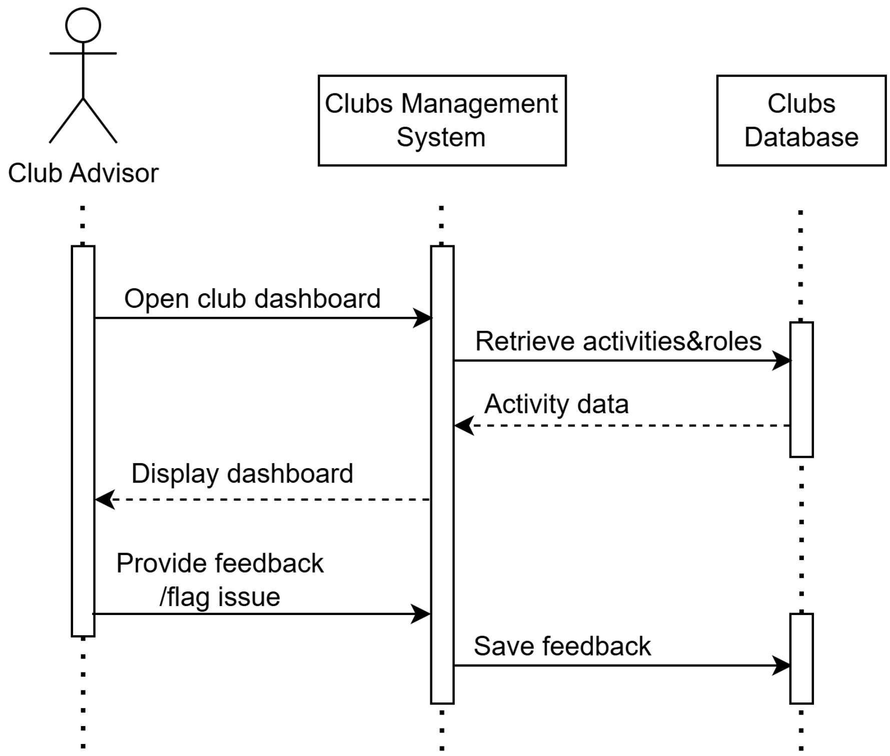

Figure 3.6.1 Sequence Diagram for Monitor Club Activities

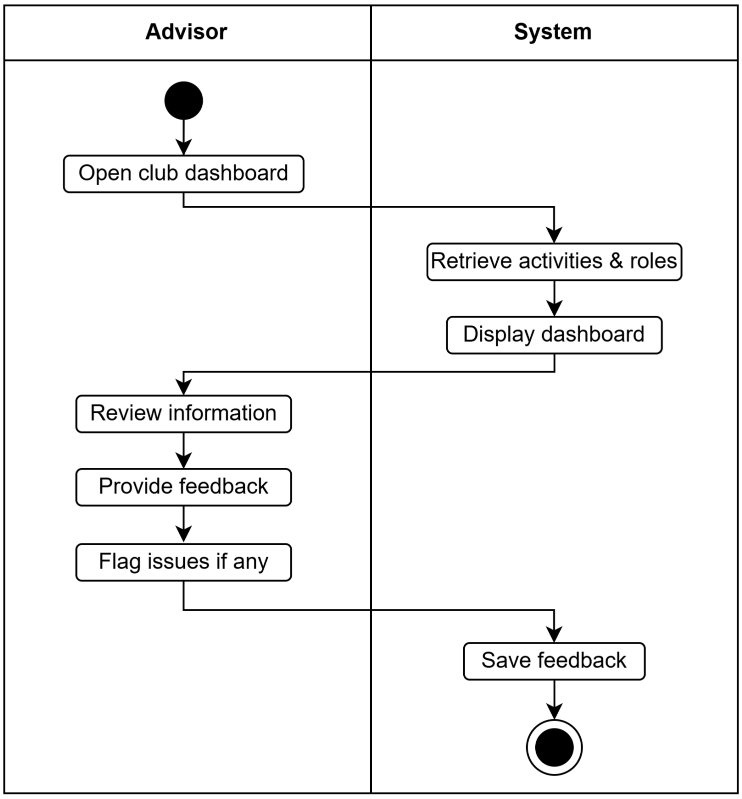

Figure 3.6.2 Activity Diagram for Monitor Club Activities

4.6.2.7 US07 Monitor Registered Clubs

Table 4.7 shows the user story description, followed by the sequence diagram in Figure 3.7.1 and the activity diagram in Figure 3.7.2.

Table 4.7 User Story Description for Monitor Registered Clubs

User story: Monitor Registered Clubs
ID: US 07
User Story Description: **As a** Student Affairs Officer (HEP) **I want** to monitor all registered clubs **So that** university clubs remain compliant and active.
Flow of events: 1. HEP views club information. 2. Reviews status, members, activities.
Alternative flow: 1. If there are no registered clubs, it will be an empty list.
Acceptance Criteria: Precondition: Clubs exist in the system. Postcondition: Club information is updated.
Exception flow: Database access error.

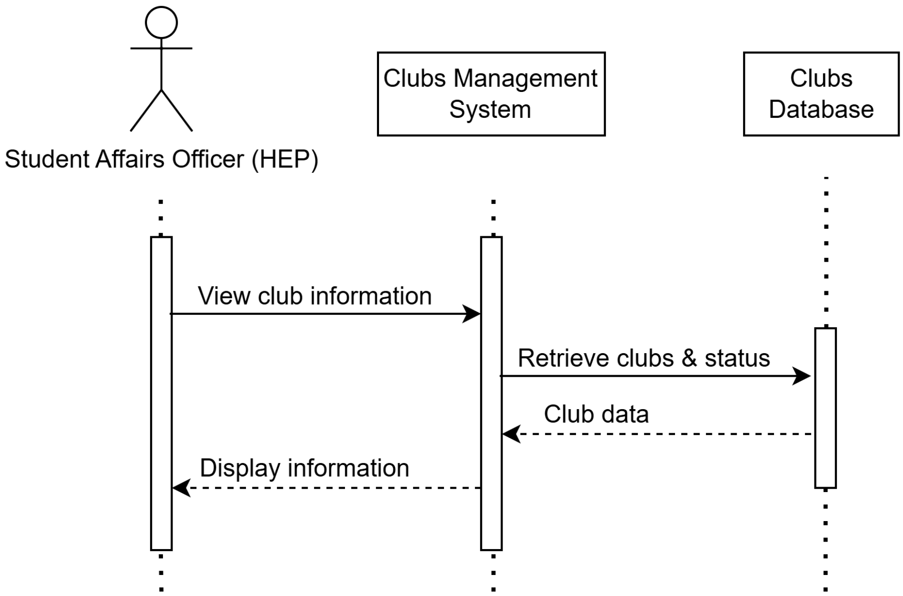

Figure 3.7.1 Sequence Diagram for Monitor Registered Clubs

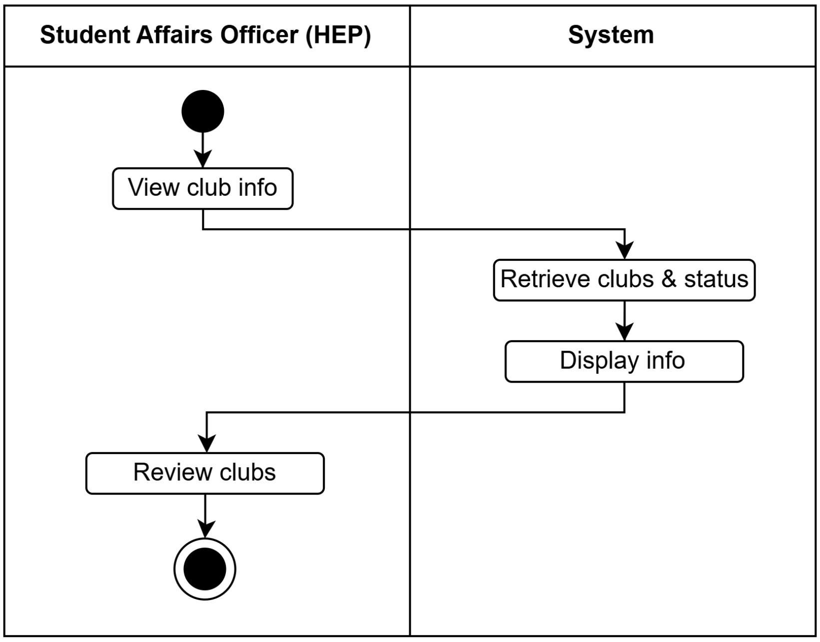

Figure 3.7.2 Activity Diagram for Monitor Registered Clubs

4.6.2.8 US08 Approve Club

Table 4.8 shows the user story description, followed by the sequence diagram in Figure 3.8.1 and the activity diagram in Figure 3.8.2.

Table 4.8 User Story Description for Approve Club

User story: Approve Club
ID: US 08
User Story Description: **As a** Student Affairs Officer (HEP) **I want** to approve or reject club registration **So that** the club's official status is determined.
Flow of events: 1. HEP reviews documents. 2. HEP approves or rejects. 3. System updates status and notifies committee.
Alternative flow: 1. Proposal returned for correction.
Acceptance Criteria: Precondition: Registration has been reviewed. Postcondition: Club becomes registered or rejected.
Exception flow: System fails to update status.

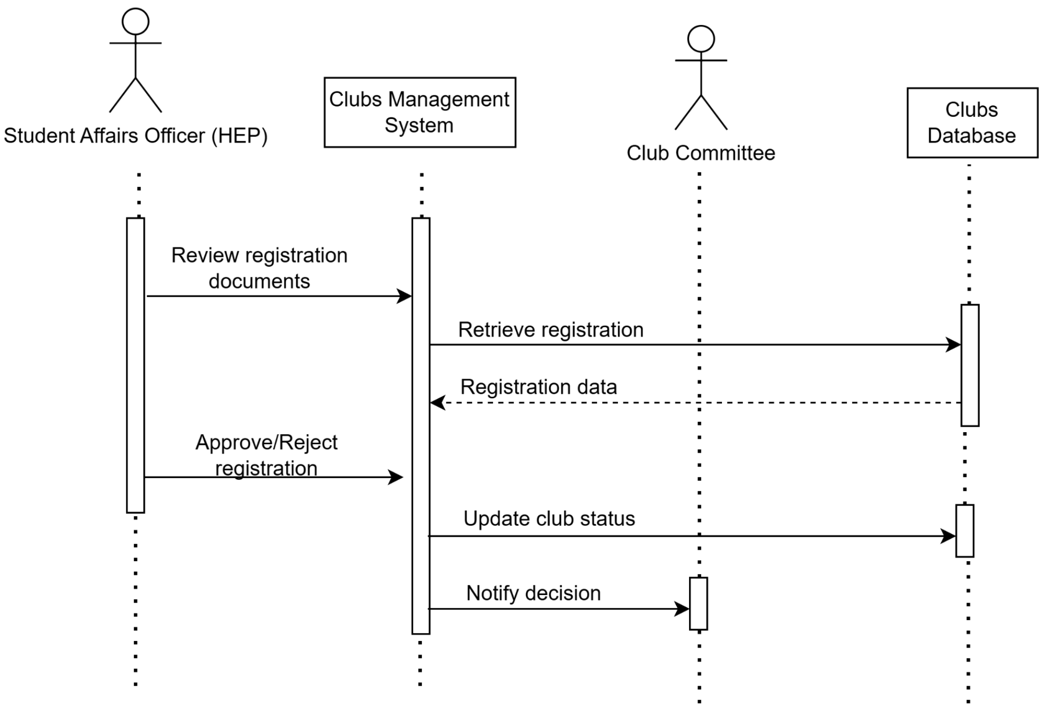

Figure 3.8.1 Sequence Diagram for Approve Club

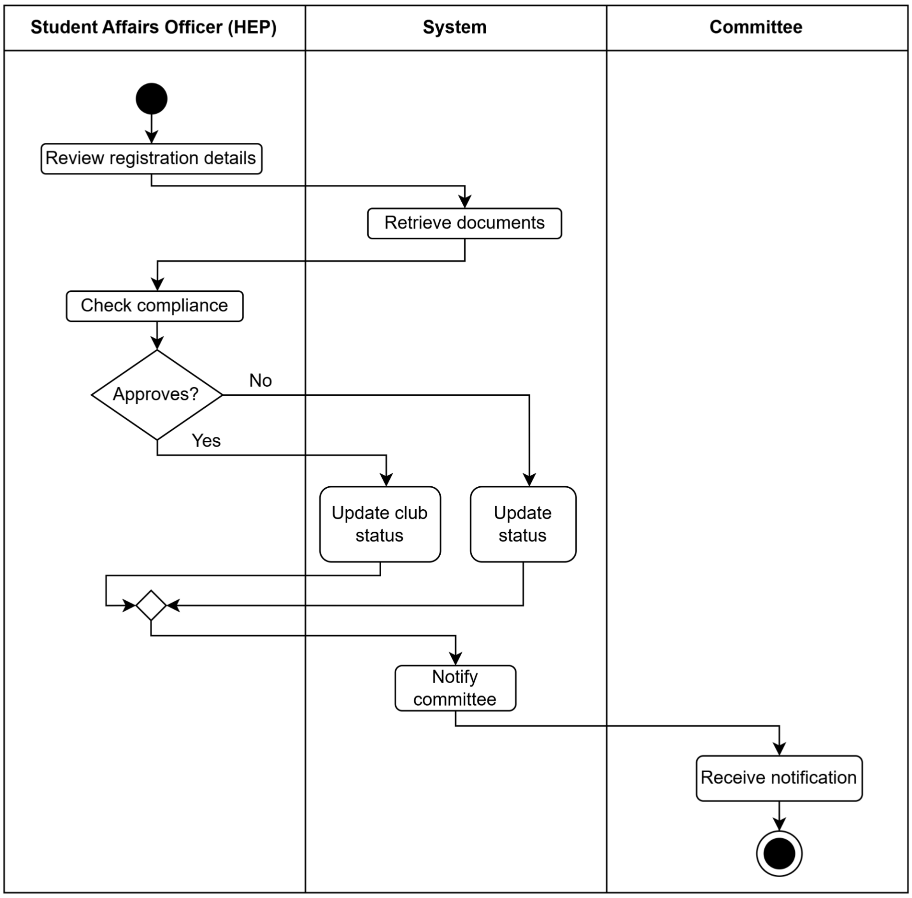

Figure 3.8.2 Activity Diagram for Approve Club

4.6.2.9 US09 Submit Event Proposal

Table 4.9 shows the user story description, followed by the sequence diagram in Figure 3.9.1 and the activity diagram in Figure 3.9.2.

Table 4.9 User Story Description for Submit Event Proposal

User story: Submit Event Proposal
ID: US09
User Story Description: **As a** Club Committee **I want** submit an event proposal **So that** the proposal can be reviewed and approved by the Club Advisor
Flow of events: 1. The club committee logs into the system. 2. The system displays the dashboard. 3. The club committee selects “Submit Event Proposal”. 4. The club committee fills in event proposal details and uploads supporting documents. 5. The club committee submits the proposal. 6. The system validates the data. 7. The system saves the proposal with “Pending” status. 8. The system sends notifications to the club advisor 9. The system displays confirmation messages.
Alternative flow: 1. If required information is incomplete, the system prompts the user to complete the form.
Acceptance Criteria: Preconditions: 1. The club committee is logged into the system. Postcondition: 1. The event proposal is successfully saved in the system. 2. Proposal status is set to “Pending”. 3. A notification email is sent to the club advisor.
Exception flow: 1. If the file upload exceeds the size limit, the system will display a warning message.

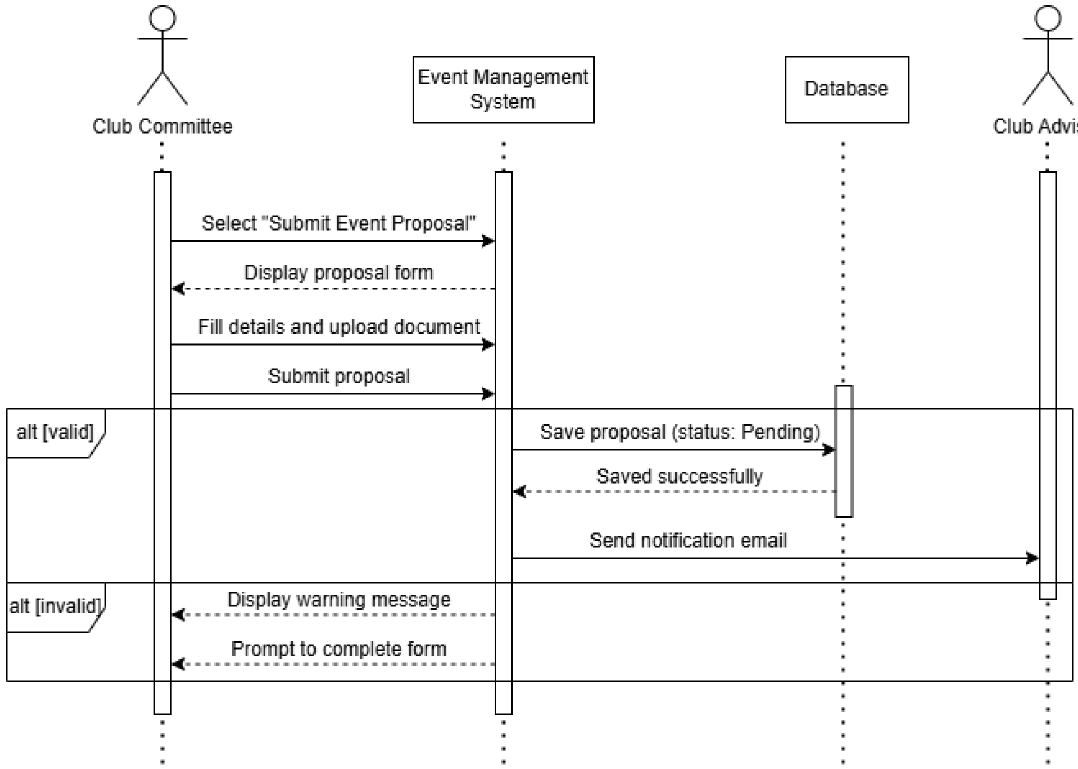

Figure 3.9.1 Sequence Diagram for Submit Event Proposal

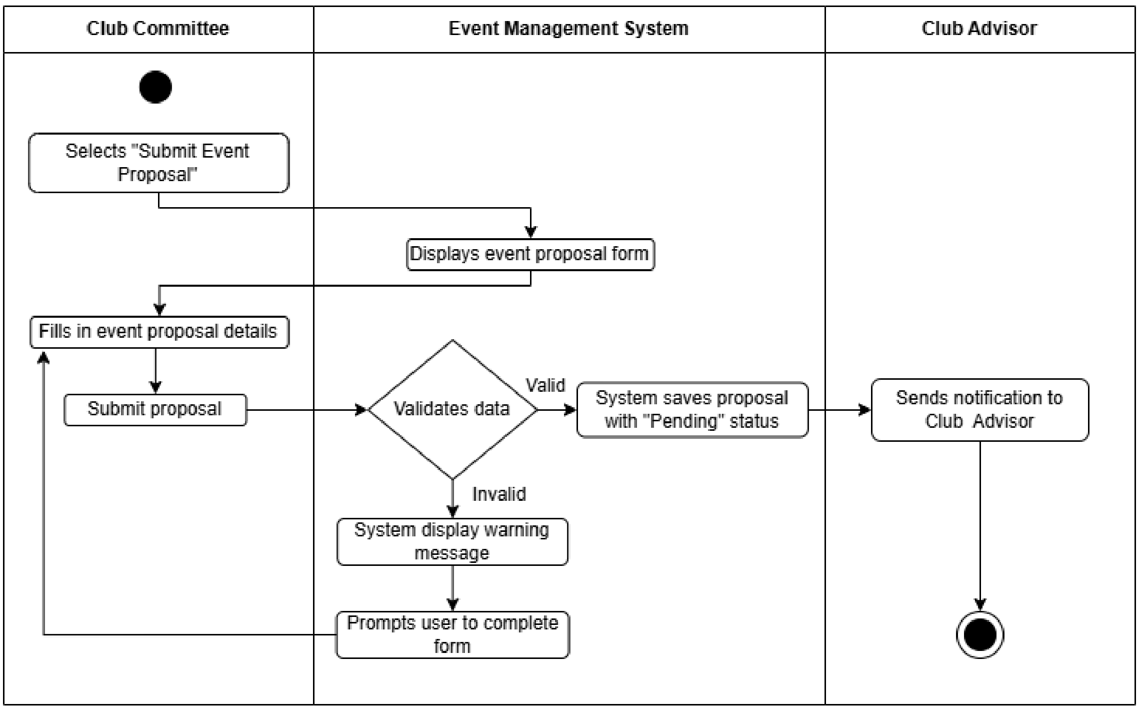

Figure 3.9.2 Activity Diagram for Submit Event Proposal

4.6.2.10 US10 Manage Event Proposal

Table 4.10 shows the user story description, followed by the sequence diagram in Figure 3.10.1 and the activity diagram in Figure 3.10.2.

Table 4.10 User Story Description for Manage Event Proposal

User story: Manage Event Proposal
ID: US10
User Story Description: **As a** Club Advisor **I want** to review, approve or reject submitted proposals **So that** I can make informed decisions and provide feedback to club committees.
Flow of events: 1. The club advisor logs into the system. 2. The system displays the dashboard. 3. The club advisor selects “Review Proposals”. 4. If there are pending proposals, the system displays the list of proposals. 5. The club advisor selects a proposal to review. 6. If the proposal meets the criteria: a. The club advisor clicks “Approve”. b. The system prompts for optional remarks. c. The club advisor enters remarks and confirms. d. The system updates status to “Approves” and records approval details 7. Else: a. The club advisor clicks “Reject”. b. The system prompts for rejection remarks c. The club advisor enter rejection reason and confirms. d. The system updates status to “Rejected” and records rejection details. 8. The system sends notification to the club committee about the proposal status. 9. The system displays a success message.
Alternative flow: 1. If there is no pending proposal then the system will show “No pending proposals”. 2. If the club advisor attempts to approve or reject without entering remarks, the system displays “Remarks are required” error.
Acceptance Criteria: Precondition: 1. The club advisor logged in with permission. 2. Proposal must be in “Pending” status Postcondition: 1. Proposal status is updated to either “Approved” or “Rejected”. 2. Decision details are recorded with timestamp. 3. Email notification is sent to the club committee.
Exception flow: 1. If there is a data retrieval error from the database, the system will display an error message. 2. If the email notification services unavailable, the system logs the error and retries sending.

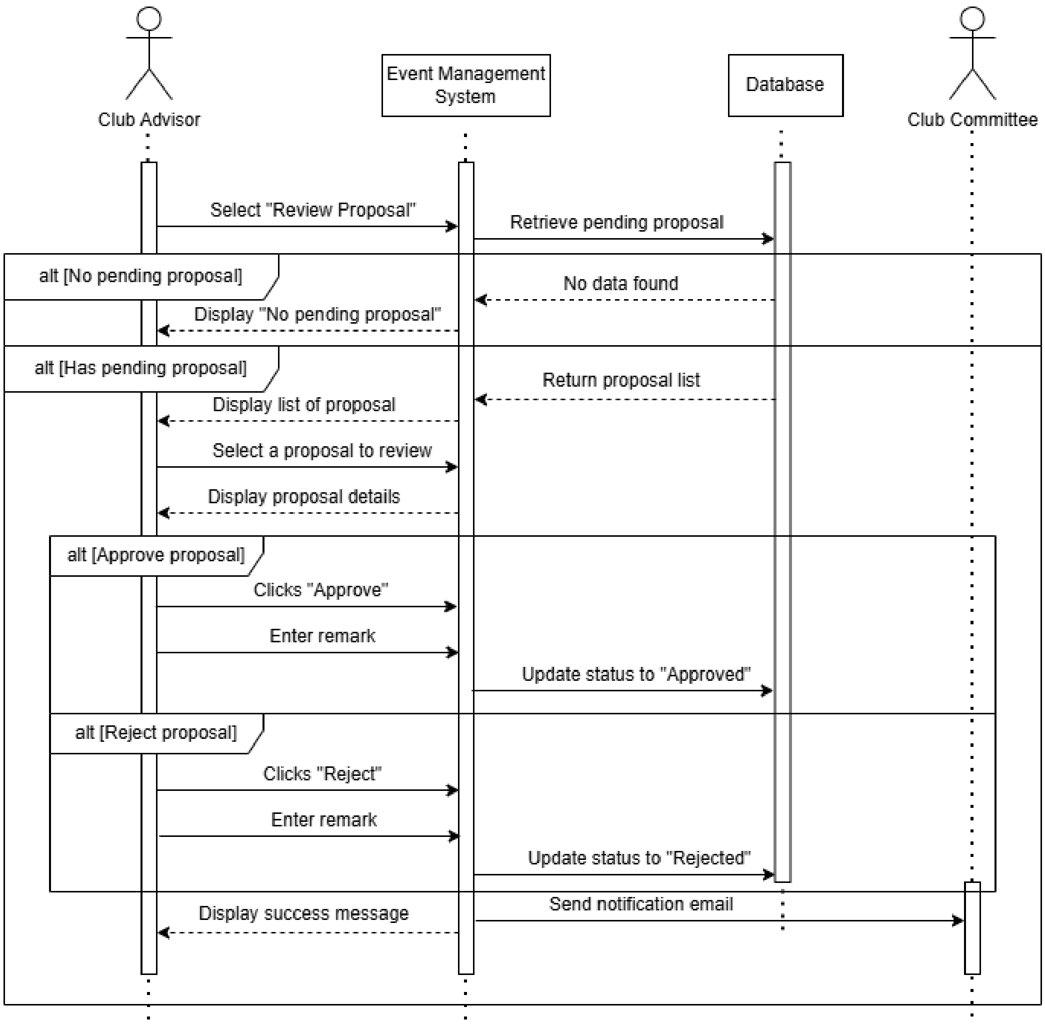

Figure 3.10.1 Sequence Diagram for Review Event Proposal

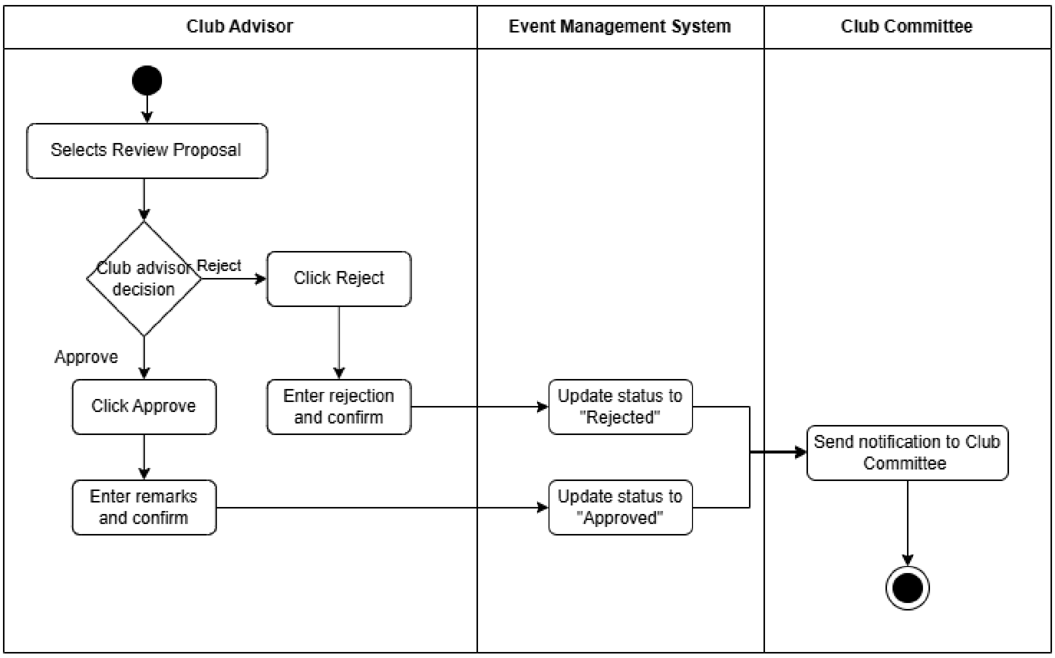

Figure 3.10.2 Activity Diagram for Review Event Proposal

4.6.2.11 US11 Track Proposal Status

Table 4.11 shows the user story description, followed by the sequence diagram in Figure 3.11.1 and the activity diagram in Figure 3.11.2.

Table 4.11 User Story Description for Track Proposal Status

User story:
ID: US11
User Story Description: **As a** Club Committee **I want** to track the status of my submitted proposals **So that** I can stay informed about the review progress and decisions.
Flow of events: 1. The club committee logs into the system. 2. The system displays the dashboard. 3. The system retrieves all proposals by this committee. 4. The system displays a list of proposals. 5. The club committee selects a specific proposal to view details. 6. The system displays detailed status, history and advisor remarks.
Alternative flow: 1. If no proposals have been submitted, the system displays "No proposals found" message. 2. The club committee can filter or search proposals in the system. 3. If the status of the proposal is "Rejected", the club committee can make revisions and resubmit the proposal.
Acceptance Criteria: Precondition: 1. Club committee must be logged in. Postcondition: 1. All proposal statuses are displayed. 2. Status history is complete with timestamps. 3. Advisor remarks are visible when available.
Exception flow: 1. If there is a data retrieval error, the system will display an error message. 2. If the status history is corrupted, the system will display a warning message.

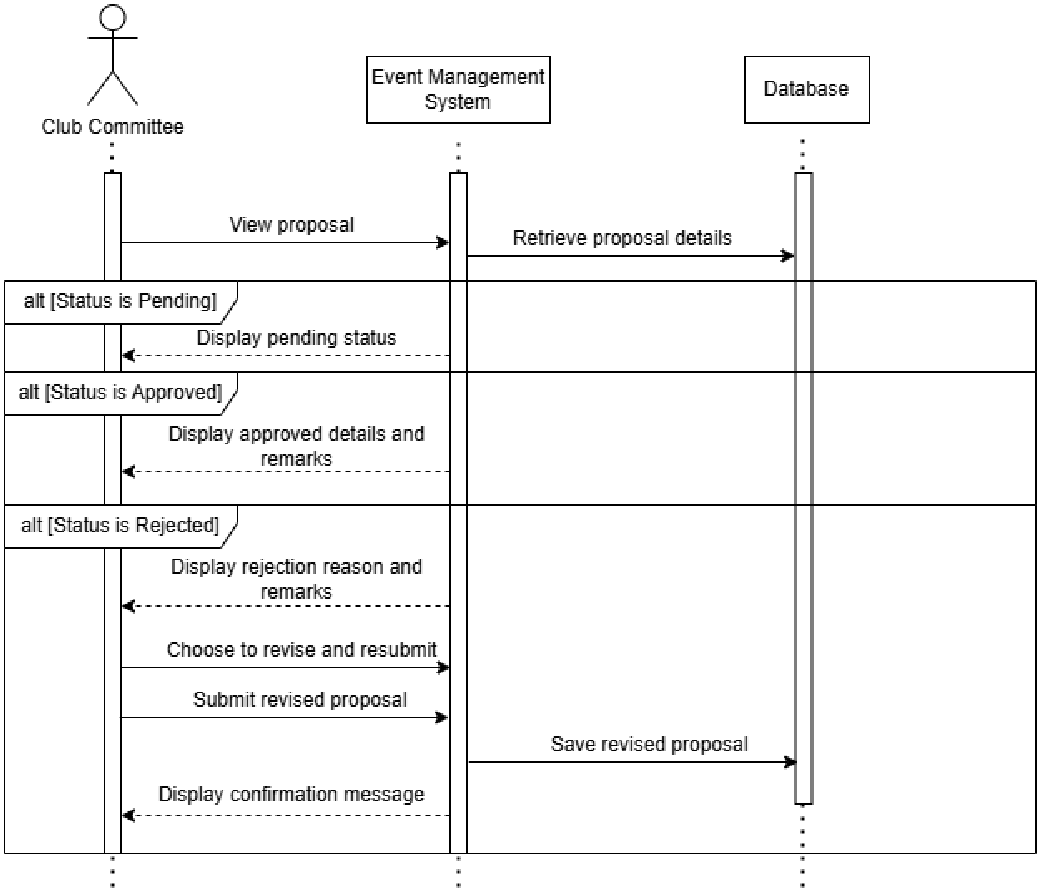

Figure 3.11.1 Sequence Diagram for Track Proposal Status

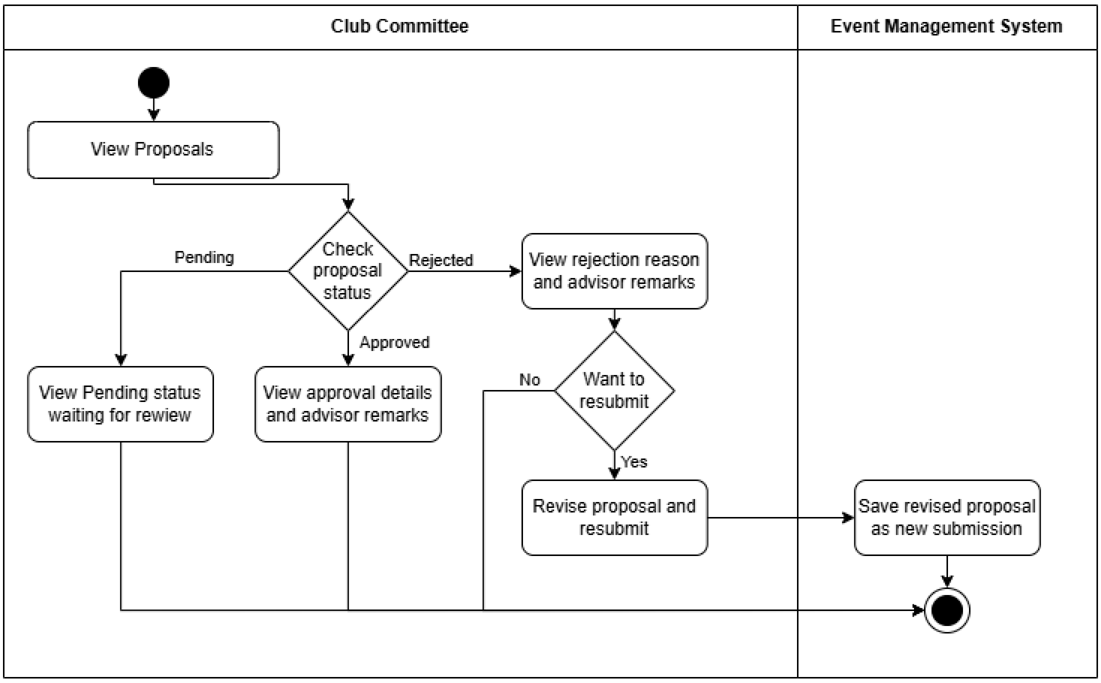

Figure 3.11.2 Activity Diagram for Track Proposal Status

4.6.2.12 US12 Register Club Events

Table 4.12 shows the user story description, followed by the sequence diagram in Figure 3.12.1 and the activity diagram in Figure 3.12.2.

Table 4.12 User Story Description for Register Club Events

User story: Register Club Events
ID: US12
User Story Description: **As a** Student **I want** to register an event **So that** I can participate in the event
Flow of events: 1. The student logs into the system. 2. The student selects "Events" 3. The system retrieves and displays brief information of the events (name, description, date, time, venue, organiser and fee) from the events data. 4. The student selects an event. 5. The system displays the details of the events 6. The user clicks the "Register" button 7. The system checks the eligibility of the user. 8. The student submits a registration form. 9. The system records the registration form. 10. The system displays a confirmation message and displays a QR code indicating that the registration has been successfully submitted.
Alternative flow: 1. If a student is not eligible to participate in the event, the system displays the requirements of the event. 2. If the event is charged, the student has to pay the registration fee. 3. If a student cancels the application, there is no request stored.
Acceptance Criteria: Precondition: 1. The student must be logged in to register for an event. 2. The system records only one registration per event. Postcondition: 1. A confirmation message and QR code is shown upon successful registration
Exception flow: 1. If a student has already registered previously or the student is not eligible for the event, the system will display a warning. 2. If there is another event registered that clashes with the event, the system will display a warning.

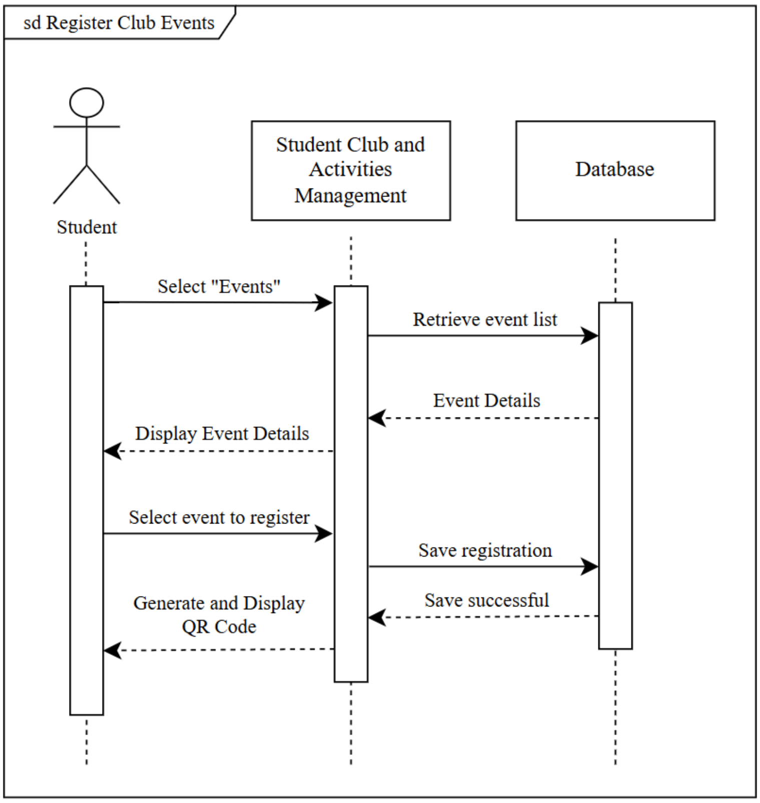

Figure 3.12.1 Sequence Diagram for Register Club Events

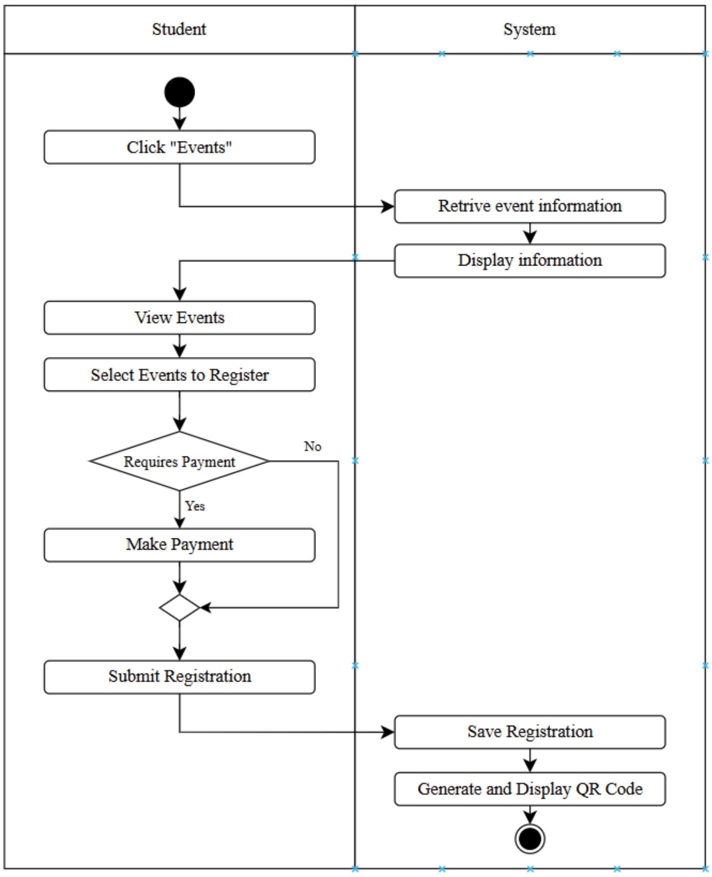

Figure 3.12.2 Activity Diagram for Register Club Events

4.6.2.13 US13 View Event Registration

Table 4.13 shows the user story description, followed by the sequence diagram in Figure 3.13.1 and the activity diagram in Figure 3.13.2.

Table 4.13 User Story Description for View Event Registration

User story: View Event Registration
ID: US13
User Story Description: **As a** Club Committee **I want** to view the registered participants list **So that** the number of participants can be traced from time to time
Flow of events: 1. The club committee logs into the system. 2. The club committee selects "Events" 3. The club committee retrieves and displays events (name, description, date, time, venue, organiser and fee) 4. The club committee selects an event. 5. The system checks if the user is the committee of the event 6. The system displays the details of the events with an additional view details button 7. The club committee selects to view additional details 8. The system displays the detailed information of the event. 9. The club committee views the participants registered in the event.
Alternative flow: 1. If there is no participants, system displays "No participants registered"
Acceptance Criteria: Precondition: 1. The club committee is still in service and the event is organised by the club that the club committee is in service. Postcondition: 1. Club committee views the registered participants
Exception flow: 1. If an error occurs, the system pops up a message "Unable to load events."

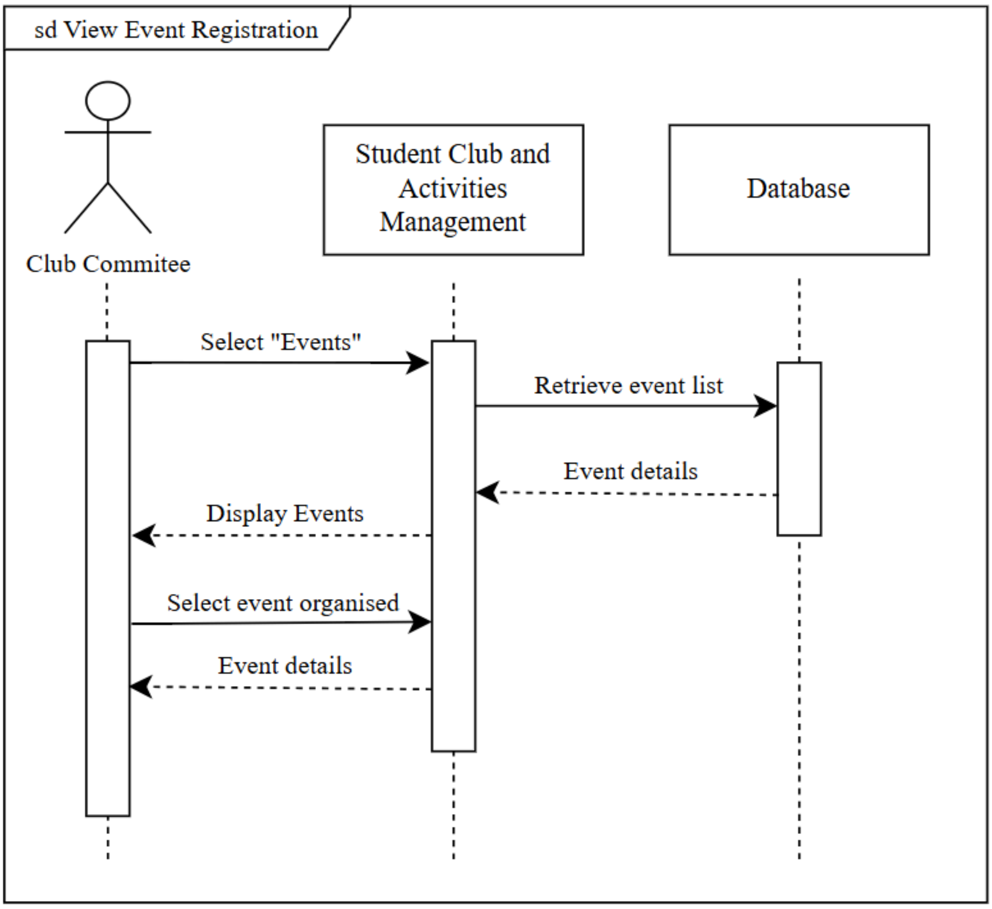

Figure 3.13.1 Sequence Diagram for View Event Registration

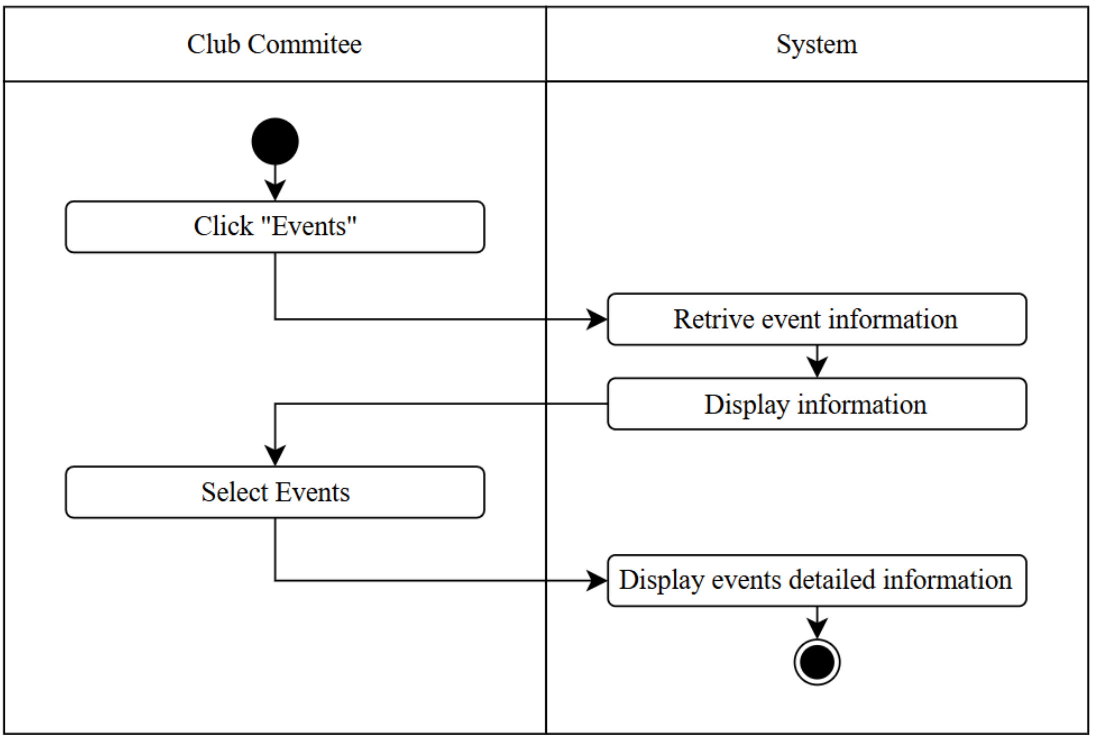

Figure 3.13.2 Activity Diagram for View Event Registration

4.6.2.14 US14 Scan Attendance QR

Table 4.21 shows the user story description, followed by the sequence diagram in Figure 3.14.1 and the activity diagram in Figure 3.14.2.

Table 4.14 User Story Description for Scan Attendance QR

User story: Scan Attendance QR
ID: US14
User Story Description: **As a** Student **I want** to scan attendance QR **So that** my attendance in this event can be recorded.
Flow of events: 1. The student logs into the system. 2. The student selects the option to scan the attendance QR code. 3. The system activates the device camera. 4. The student scans the QR code displayed at the event venue. 5. The system validates student information and QR code against the event information. 6. The system records the attendance with the attendance recording module. 7. The system displays a confirmation message indicating that attendance has been successfully submitted.
Alternative flow: 1. If the QR code has already been scanned by the student for the same event, the system informs the student that attendance has already been recorded. 2. If the event attendance window is still open, the student may retry scanning.
Acceptance Criteria: Precondition: 1. The student must be logged in to scan the QR code. 2. The QR code must correspond to a valid and approved event. 3. The system records only one attendance entry per student per event. Postcondition: 1. A confirmation message is shown upon successful scanning.
Exception flow: 1. If the QR code is invalid or expired, the system displays an error message “invalid QR code”. 2. If the camera permission is denied, the system prompts the student to enable permissions to access the camera.

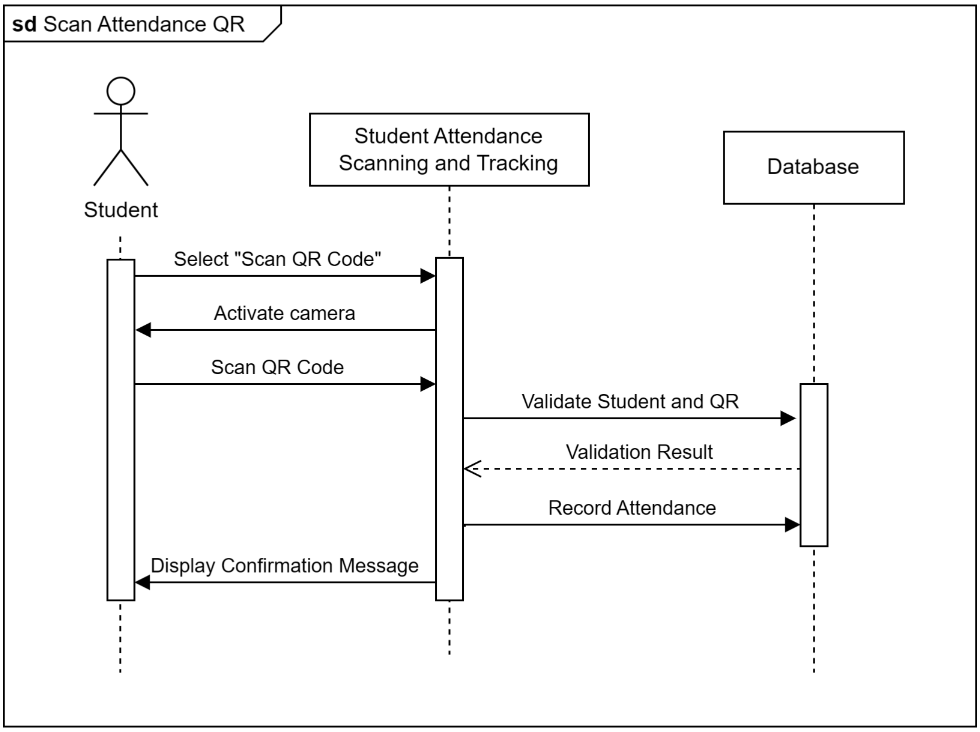

Figure 3.14.1 Sequence Diagram for Scan Attendance QR

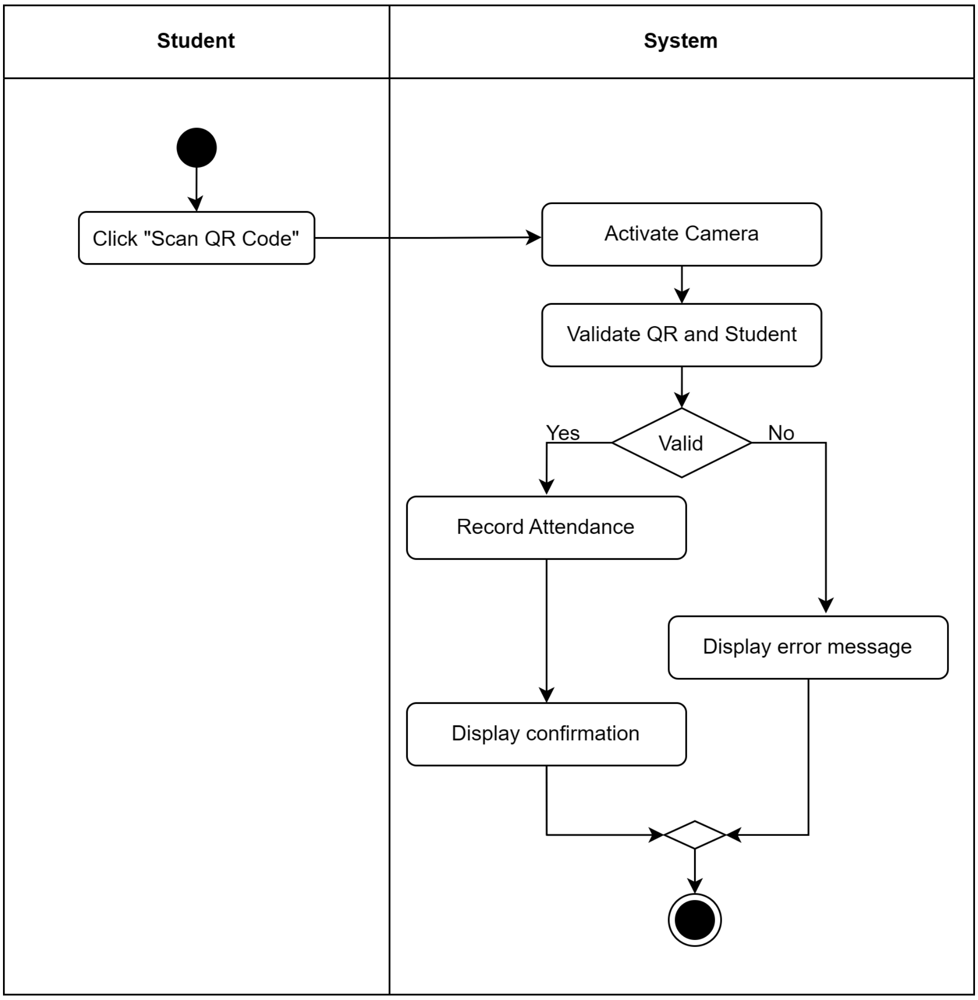

Figure 3.14.2 Activity Diagram for Scan Attendance QR

4.6.2.15 US15 Scan Student Attendance QR

Table 4.25 shows the user story description, followed by the sequence diagram in Figure 3.15.1 and the activity diagram in Figure 3.15.2.

Table 4.15 User Story Description for Scan Attendance QR

User story: Scan Student Attendance QR
ID: US15
User Story Description: **As a** Club Committee **I want** to scan student attendance QR **So that** I can record student attendance while verifying their identity.
Flow of events: 1. The club committee logs in to the system. 2. The club committee selects the option to scan student attendance QR code. 3. The system activates the device camera. 4. The club committee scans the QR code displayed by the student. 5. The system validates QR code against the registered student list. 6. The system records the attendance of the student. 7. The system displays a confirmation message indicating that attendance has been successfully submitted.
Alternative flow: 1. If the student QR code for the same event has been scanned, the system informs the club committee that this student attendance has already been recorded.
Acceptance Criteria: Precondition: 1. The club committee must be logged in to scan the QR code. 2. The QR code must belong to a student who has already registered for the corresponding event. 3. The system records only one attendance entry per student per event. Postcondition: 1. A confirmation message is shown upon successful scanning.
Exception flow: 1. If the student QR code is invalid or not for this event, the system displays an error message "QR Code Invalid or Incorrect". 2. If the camera permission is denied, the system prompts the club committee to enable permissions to access the camera.

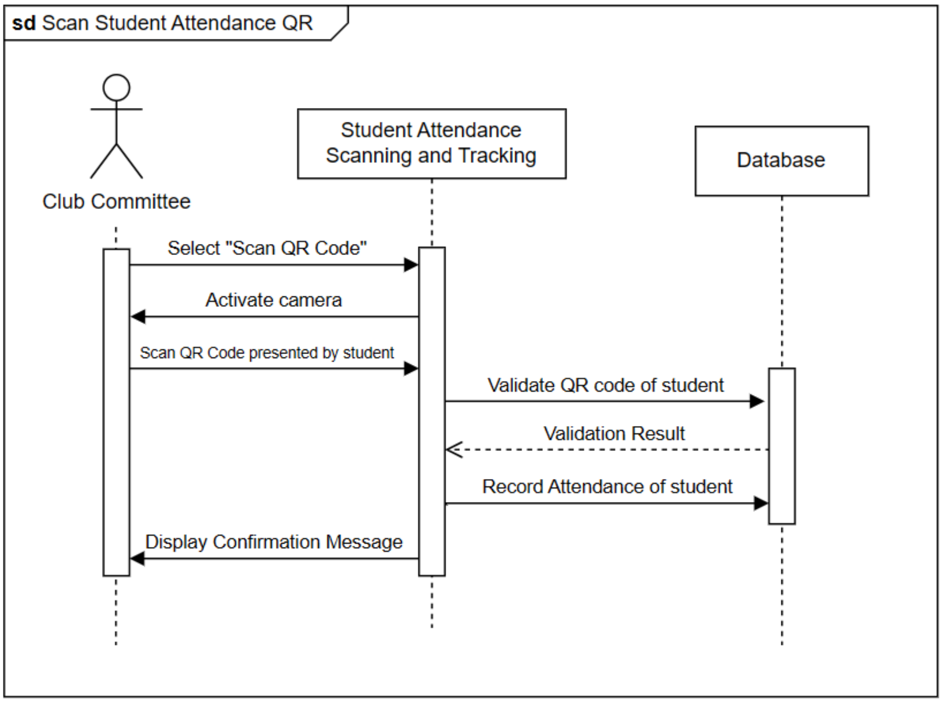

Figure 3.15.1 Sequence Diagram for Scan Student Attendance QR

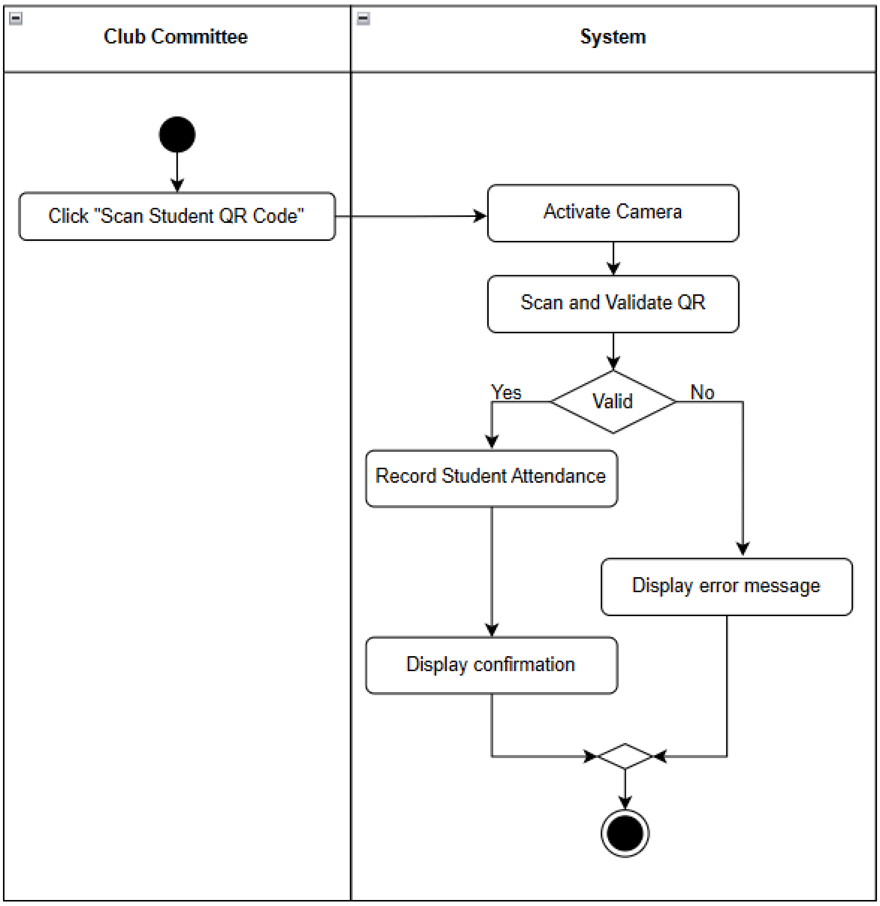

Figure 3.15.2 Activity Diagram for Scan Student Attendance QR

4.6.2.16 US16 View Attendance Summary

Table 4.16 shows the user story description, followed by the sequence diagram in Figure 3.16.1 and the activity diagram in Figure 3.16.2.

Table 4.16 User Story Description for View Attendance Summary

User story: View Attendance Summary
ID: US16
User Story Description: **As a** Club Committee/Club Advisor/Academic Advisor/Student Affairs Officer (HEP) **I want** to view event attendance summary **So that** I am able to know which student attended the event and which student did not.
Flow of events: 1. The authorized user logs into the system. 2. The user selects an event from the event list. 3. The user chooses the option to view attendance summary. 4. The system retrieves attendance records for the selected event. 5. The system displays a summary including attended participants and related information. 6. The user reviews the attendance information.
Alternative flow: 1. The user may filter attended participants by course, faculty or year of study. 2. The user may search for an individual in the attendance list with the name of the matric number.
Acceptance Criteria: Precondition: 1. Authorized roles must be logged in using valid accounts. Postcondition: 1. Attendance summaries accurately reflect recorded attendance data. 2. Data is displayed in a clear and readable format.
Exception flow: 1. If no attendance data exists, the system displays the message “No attendance data found.”. 2. If the user does not have permission, access to the summary is denied.

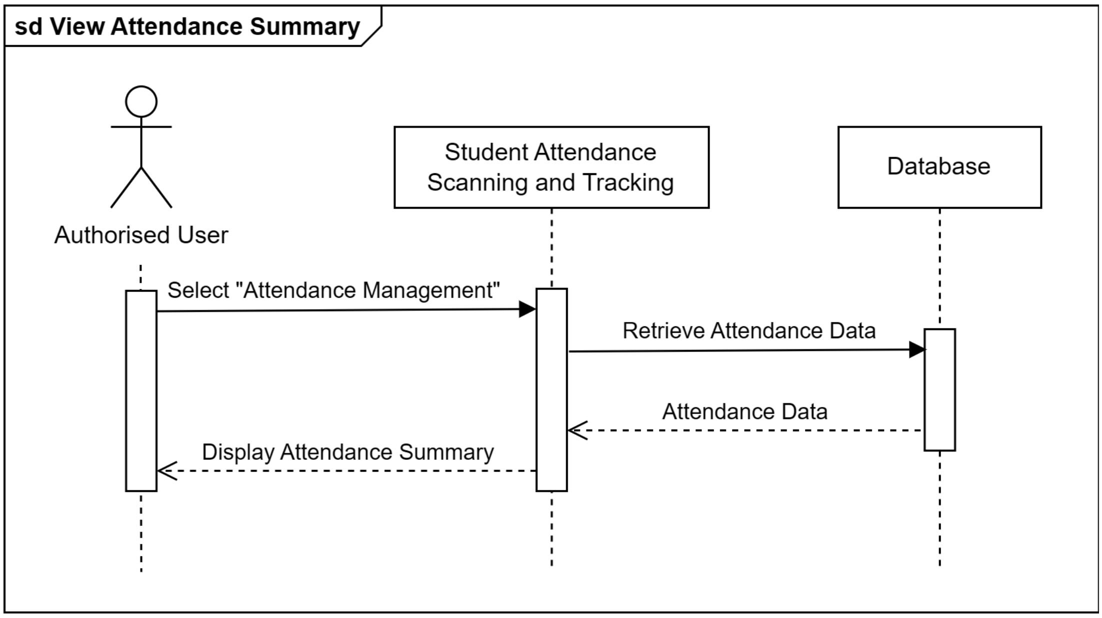

Figure 3.16.1 Sequence Diagram for View Attendance Summary

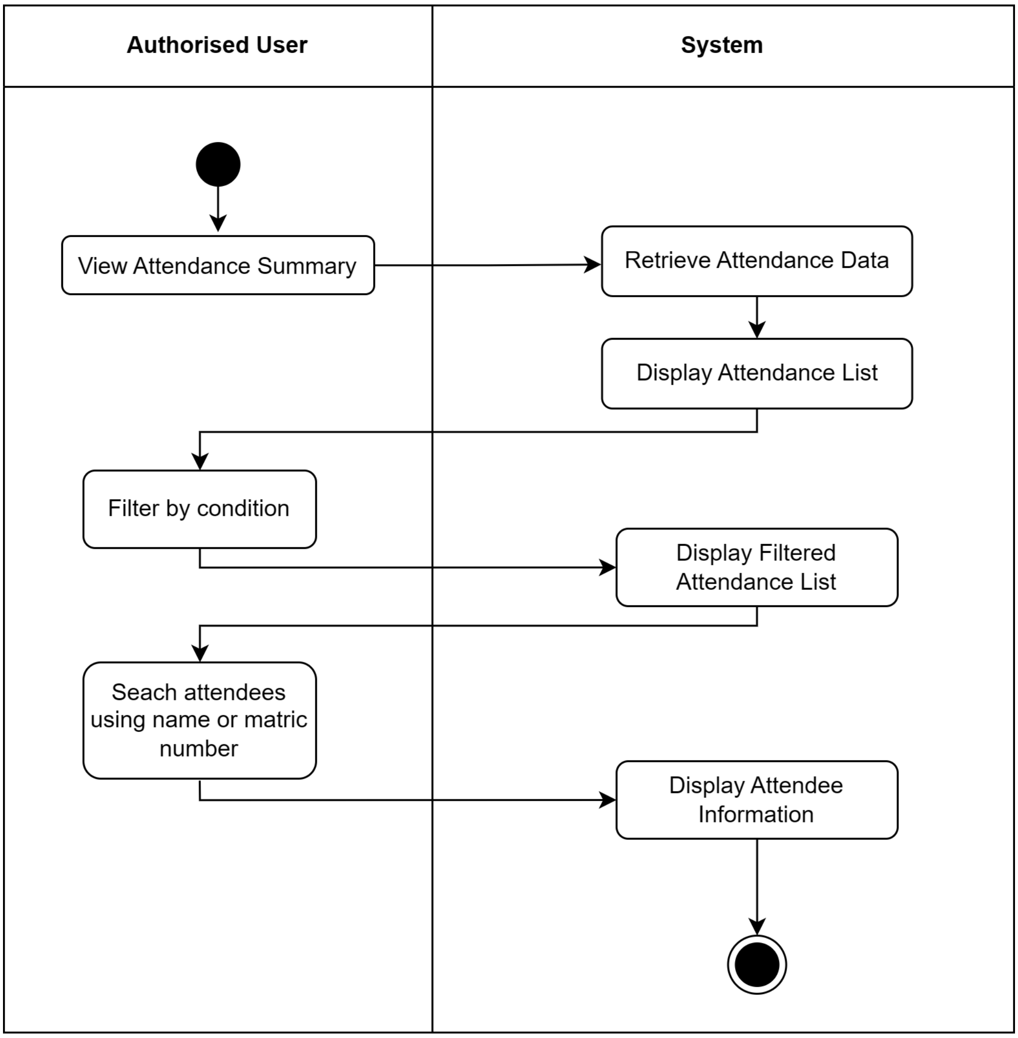

Figure 3.16.2 Activity Diagram for View Attendance Summary

4.6.2.17 US17 Apply Asset Borrowing

Table 4.17 shows the user story description, followed by the sequence diagram in Figure 3.17.1 and the activity diagram in Figure 3.17.2.

Table 4.17 User Story Description for Apply Asset Borrowing

User story: Apply Asset Borrowing
ID: US17
User Story Description: **As a** Student **I want** to apply for asset borrowing for an approved club event **So that** sufficient equipment is available to ensure the event runs smoothly.
Flow of events: 1. The student logs into the system. 2. The student selects the Borrow Asset function. 3. The student selects the Apply function. 4. The student selects an event from the list of approved events. 5. The system displays a list of available assets and their current quantity and status. 6. The student selects the required assets and specifies the quantity. 7. The student provides the reason for borrowing the assets. 8. The student submits the asset borrowing request.
Alternative flow: 1. The system displays the list of available assets. 2. Some assets have limited quantities available. 3. The student selects a quantity that is within the available limit. 4. The system allows the request to proceed with the selected quantity.
Acceptance Criteria: Precondition: 1. Students are logged into the system using a valid student account. 2. The involved event must be approved before applying for asset borrowing. 3. The system must display accurate asset availability and status. Postcondition: 1. The system must record the borrowing request with a “Pending” status upon submission. 2. The system must provide confirmation after successful submission.
Exception flow: 1. If the student selects an asset that has zero available quantity, the system displays an error message indicating that the asset is not available. 2. If the student submits the borrowing request without completing mandatory fields, the system displays validation messages.

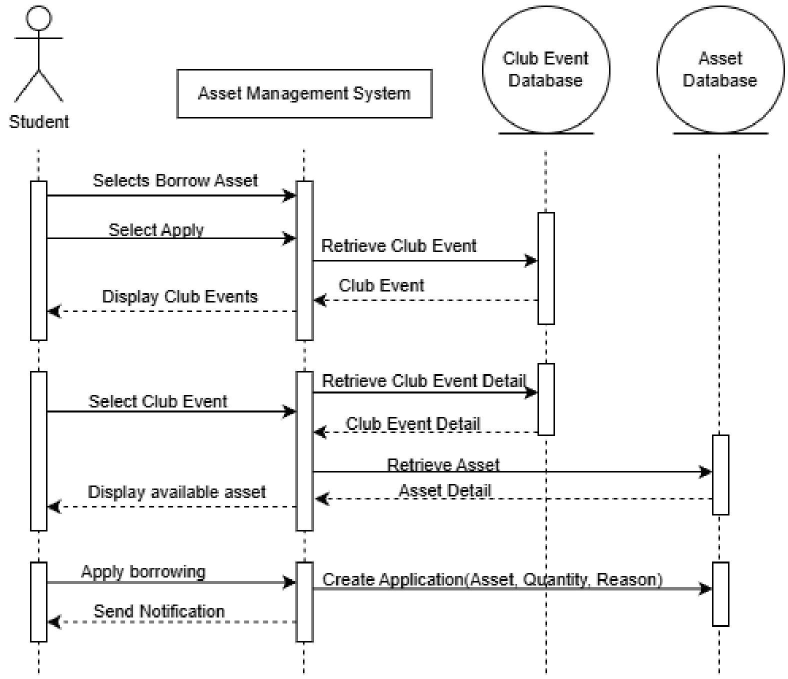

Figure 3.17.1 Sequence Diagram for Apply Asset Borrowing

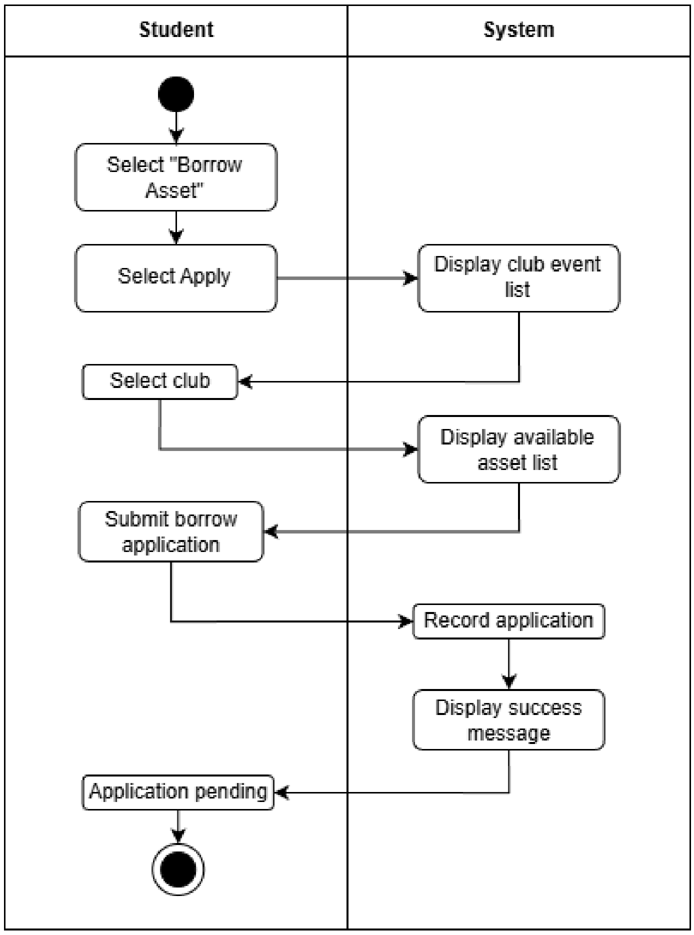

Figure 3.17.2 Activity Diagram for Apply Asset Borrowing

4.6.2.18 US18 View Borrowing Status

Table 4.18 shows the user story description, followed by the sequence diagram in Figure 3.18.1 and the activity diagram in Figure 3.18.2.

Table 4.18 User Story Description for View Borrowing Status

User story: View Borrowing Status
ID: US18
User Story Description: **As a** Student **I want** to view my asset borrowing application status **So that** I can know whether my request has been approved, rejected, or is still pending.
Flow of events: 1. The student logs into the system. 2. The student selects the Borrow Asset function. 3. The system displays the list of asset borrowing applications made by the student. 4. The student can select the application ID to view application details.
Alternative flow: 1. The student selects the Borrow Asset function. 2. The system detects that no borrowing applications exist for the student. 3. The system displays a message indicating that no asset borrowing applications are available.
Alternative flow: 1. The student views the list of borrowing applications. 2. The student filters or sorts the list by status, such as Pending, Approved, or Rejected. 3. The system updates the displayed list accordingly.
Acceptance Criteria: Precondition: 1. Students are logged into the system using a valid student account. 2. The application must be successfully submitted to be displayed. Postcondition: 1. Each application must show its application ID, submission date, and current status. 2. The system must display detailed information when an application ID is selected. 3. The application status must reflect its most recent approval decision.
Exception flow: 1. If the user does not have permission, access to the list of applications is denied.

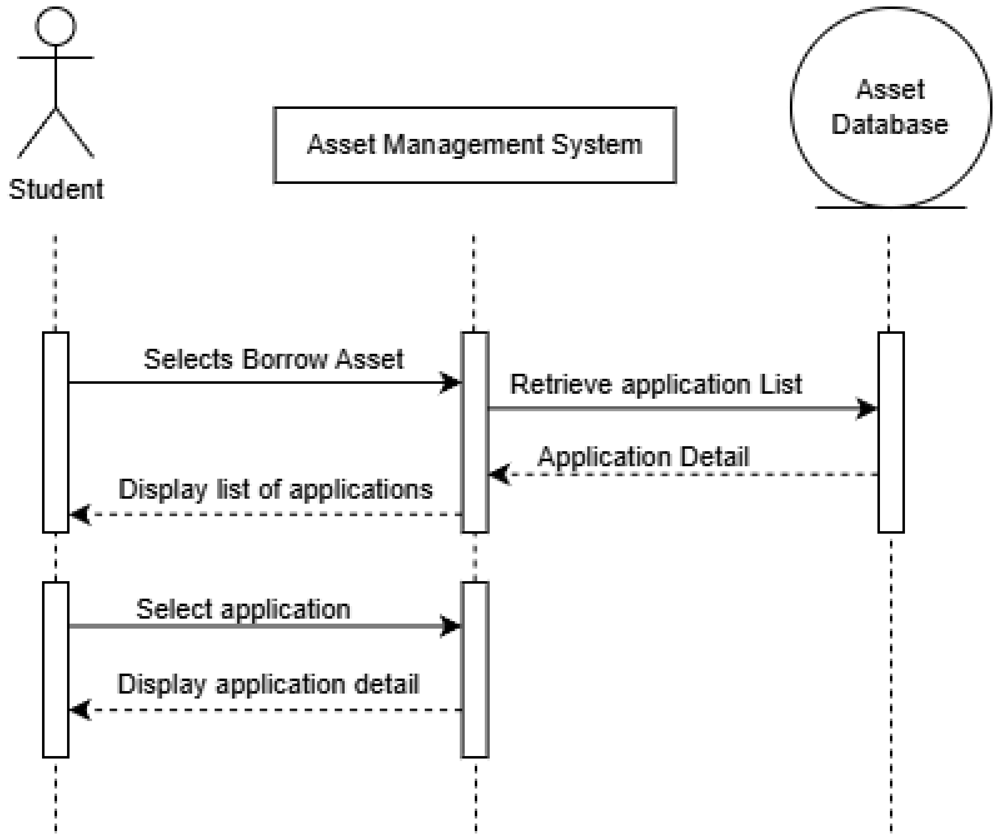

Figure 3.18.1 Sequence Diagram for View Borrowing Status

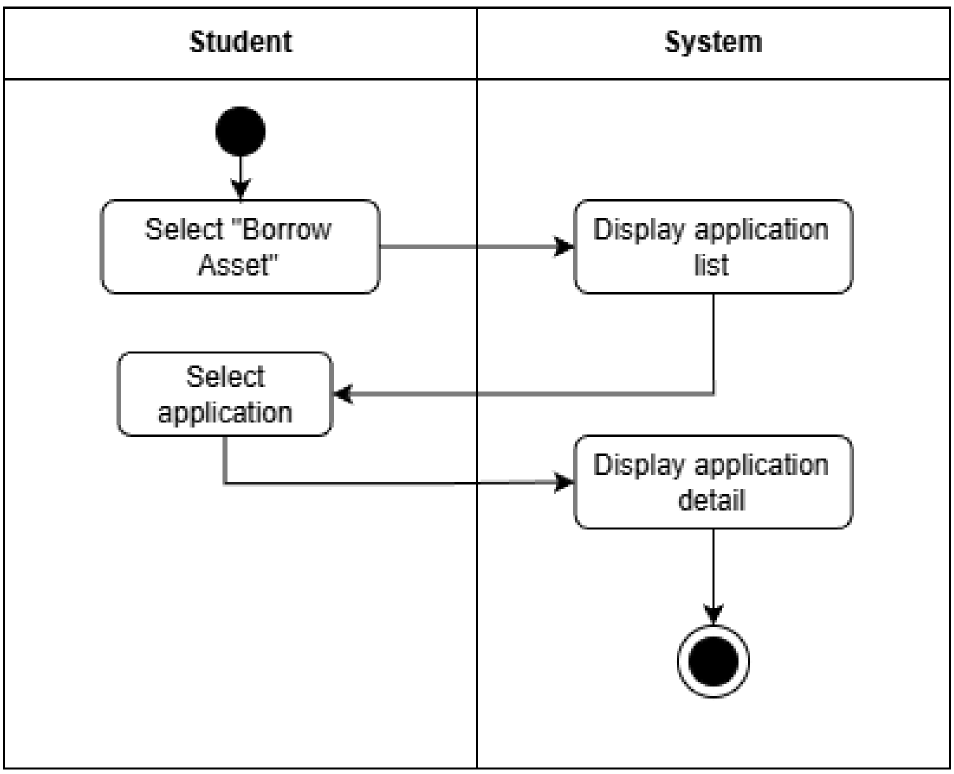

Figure 3.18.2 Activity Diagram for View Borrowing Status

4.6.2.19 US19 Return Asset

Table 4.19 shows the user story description, followed by the sequence diagram in Figure 3.19.1 and the activity diagram in Figure 3.19.2.

Table 4.19 User Story Description for Return Asset

User story: Return Asset
ID: US19
User Story Description: **As a** Student **I want** to return the borrowed assets after an event **So that** the assets can be recorded as returned.
Flow of events: 1. The student logs into the system. 2. The student selects the Borrow Asset function. 3. The system displays a list of the student's approved and active asset borrowing records. 4. The student selects the borrowing application to return assets. 5. The student selects the Return Asset function. 6. The student submits the asset return request. 7. The system records the return and updates the borrowing status to "Return Pending".
Alternative flow: 1. The student returns the assets before the expected return date. 2. The system accepts the return and updates the borrowing status to "Return Pending".
Acceptance Criteria: Precondition: 1. Students are logged into the system using a valid student account. 2. The asset borrowing application must be approved and active to display return asset function. Postcondition: 1. The system must update asset availability after assets are returned. 2. The system must record the actual return date. 3. The system must display a confirmation after successful submission.
Exception flow: 1. If the user does not have permission, access to the application details is denied.

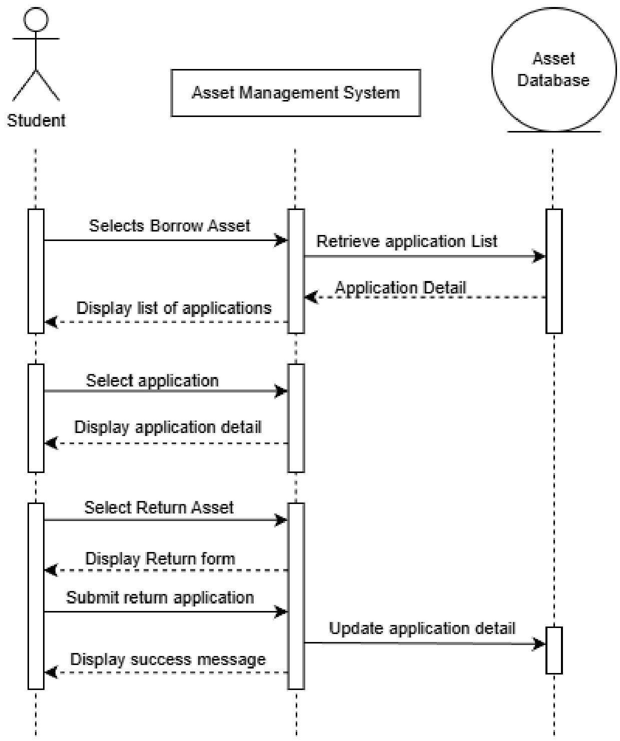

Figure 3.19.1 Sequence Diagram for Return Asset

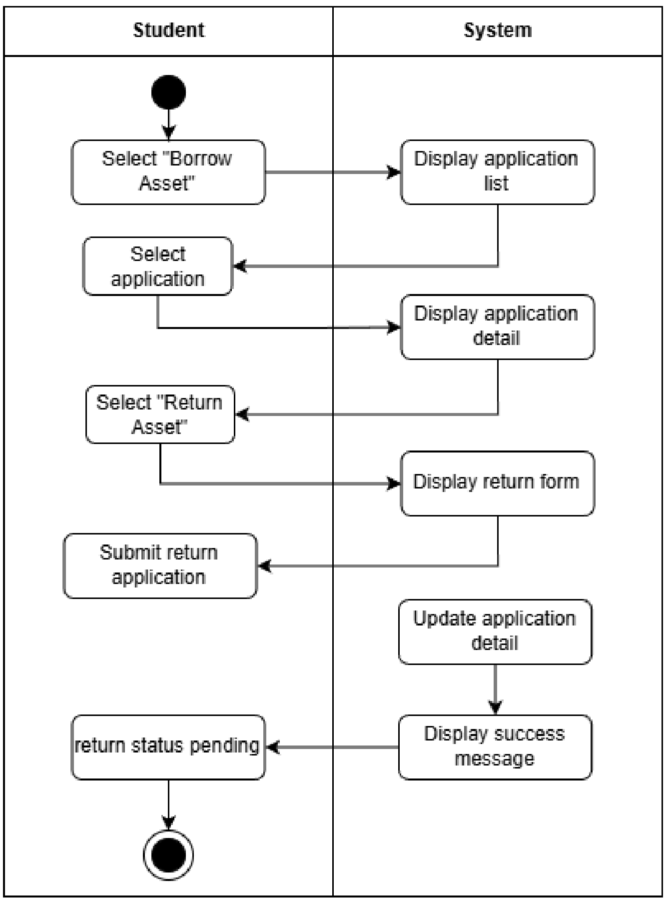

Figure 3.19.2 Activity Diagram for Return Asset

4.6.2.20 US20 Review Asset Borrowing Application

Table 4.20 shows the user story description, followed by the sequence diagram in Figure 3.20.1 and the activity diagram in Figure 3.20.2.

Table 4.20 User Story Description for Review Asset Borrowing Application

User story: Review Asset Borrowing Application
ID: US20
User Story Description: **As a** Student Affairs Office (HEP) **I want** to review asset borrowing application **So that** I can review and approve or reject asset borrowing application
Flow of events: 1. The Student Affairs Office logs into the system. 2. The Student Affairs Office selects the Asset function. 3. The Student Affairs Office selects the Borrow Application function. 4. The system displayed a list of asset borrow applications submitted. 5. The Student Affairs Office selects an application to view its details. 6. The Student Affairs Office reviews the application information. 7. The Student Affairs Office approves the application. 8. The system updates the application status to Approved and notifies the student.
Alternative flow: 1. The Student Affairs Office reviews the application information. 2. The Student Affairs Office rejects the application. 3. The Student Affairs Office gives a rejection reason (Optional). 4. The system updates the application status to Rejected and notifies the student.
Acceptance Criteria: Precondition: 1. The Student Affairs Office must be authenticated before accessing the review function. 2. At least one asset borrowing application has been successfully submitted. Postcondition: 1. The application status is updated to Approved or Rejected. 2. The student is notified of the application outcome.
Exception flow: 1. If the user is not authenticated, access to the list of applications is denied.

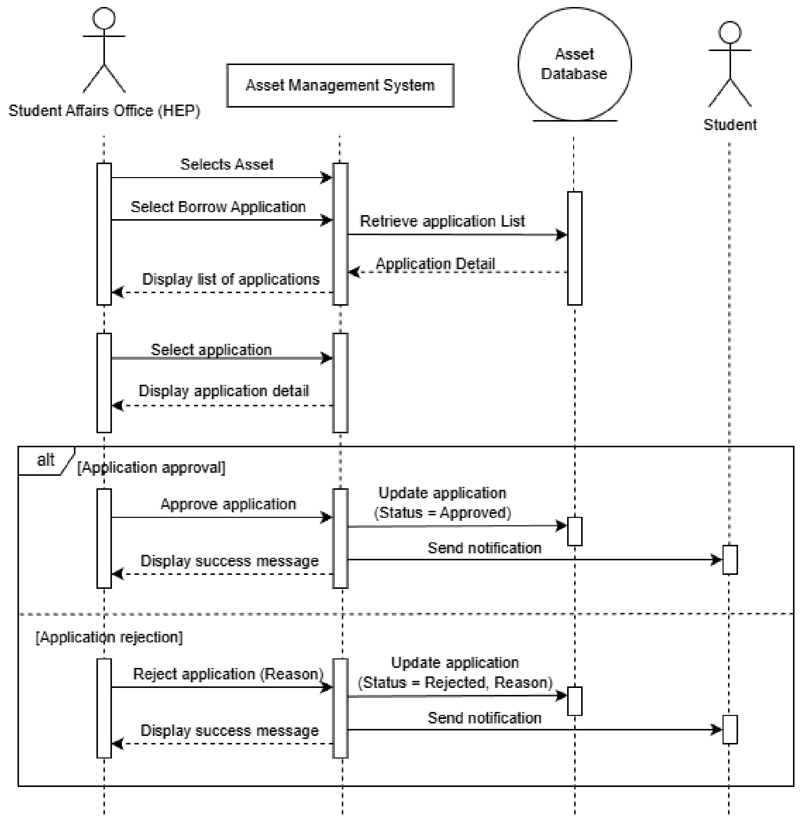

Figure 3.20.1 Sequence Diagram for Review Asset Borrowing Application

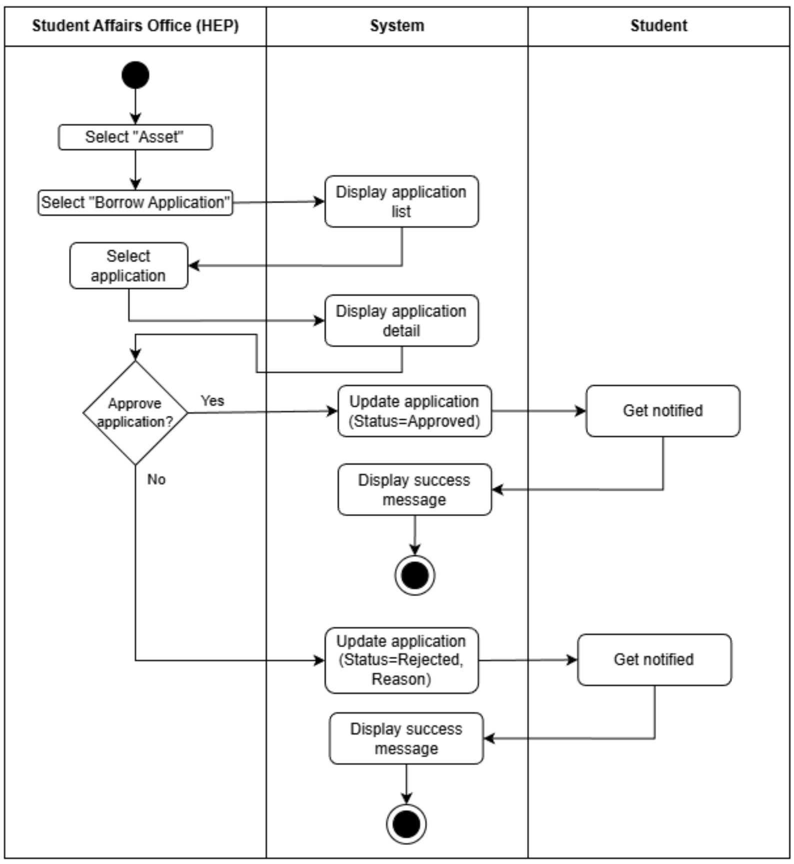

Figure 3.20.2 Activity Diagram for Review Asset Borrowing Application

4.6.2.21 US21 Manage Asset

Table 4.21 shows the user story description, followed by the sequence diagram in Figure 3.21.1 and the activity diagram in Figure 3.21.2.

Table 4.21 User Story Description for Manage Asset

User story: Manage Asset
ID: US21
User Story Description: **As a** Student Affairs Office (HEP) **I want** to manage assets in the system **So that** asset information is accurate and up to date for borrowing by student clubs.
Flow of events: 1. The Student Affairs Office logs into the system. 2. The Student Affairs Office selects the Asset function. 3. The Student Affairs Office selects the Manage Asset function. 4. The system displays a list of existing assets. 5. The Student Affairs Office selects an asset to manage. 6. The Student Affairs Office updates asset information such as asset name, quantity, condition, or availability status. 7. The Student Affairs Office saves the changes. 8. The system updates the asset information and confirms the successful update.
Alternative flow: 1. The Student Affairs Office selects Add New Asset. 2. The Student Affairs Office enters the asset details. 3. The Student Affairs Office saves the asset information. 4. The system records the new asset and displays a confirmation message.
Alternative flow: 1. The Student Affairs Office selects Remove Asset. 2. The system prompts for confirmation. 3. The Student Affairs Office confirms the removal. 4. The system removes the asset from the active asset list.
Acceptance Criteria: Precondition: 1. The Student Affairs Office must be authenticated before accessing the manage function. Postcondition: 1. Asset information is updated, added, or removed successfully. 2. The system reflects the latest asset data.

Exception flow:

1. The Student Affairs Office attempts to remove an asset that is currently borrowed or linked to an active application, the system prevents the action and displays a notification.
2. The Student Affairs Office enters incomplete or invalid asset information, the system displays an error message and requests correction.

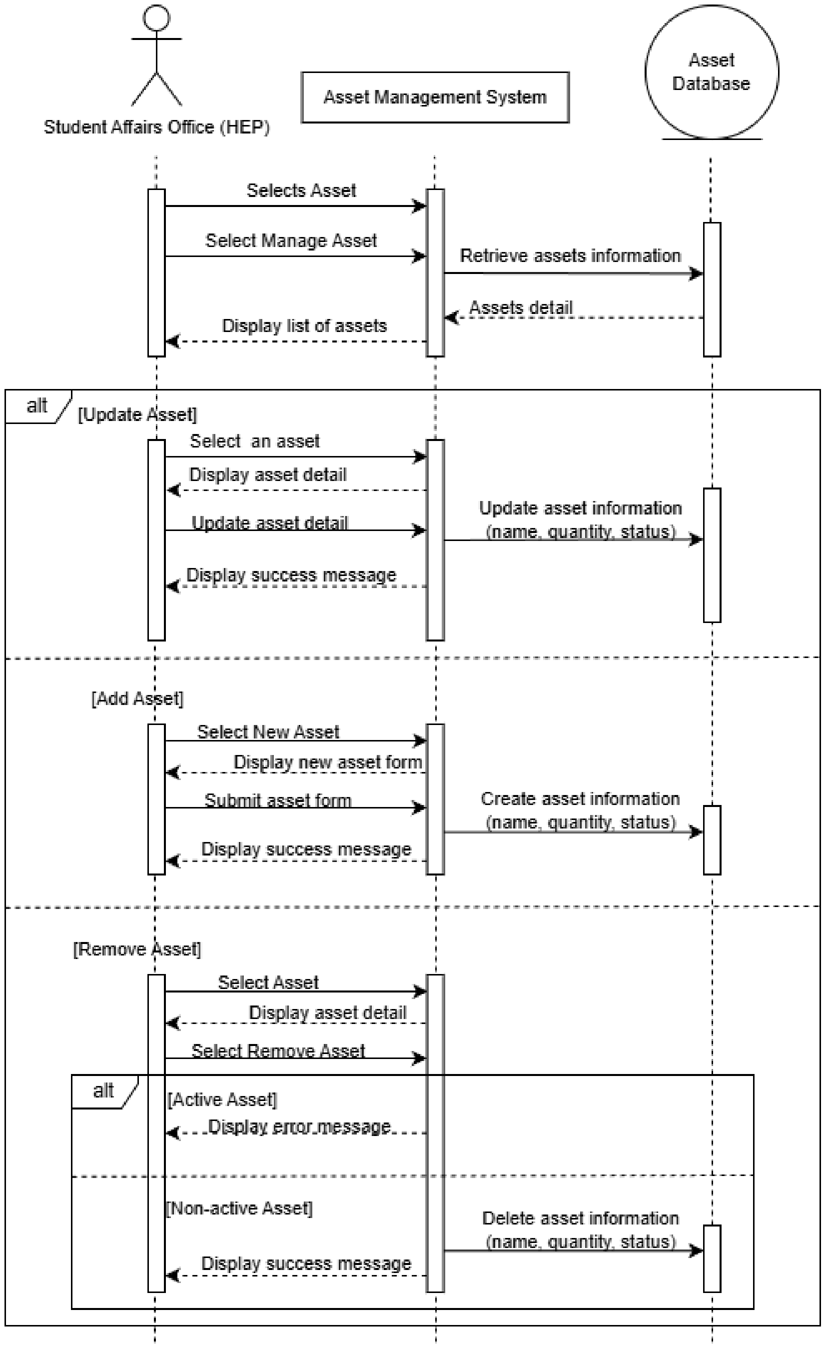

Figure 3.21.1 Sequence Diagram for Manage Asset

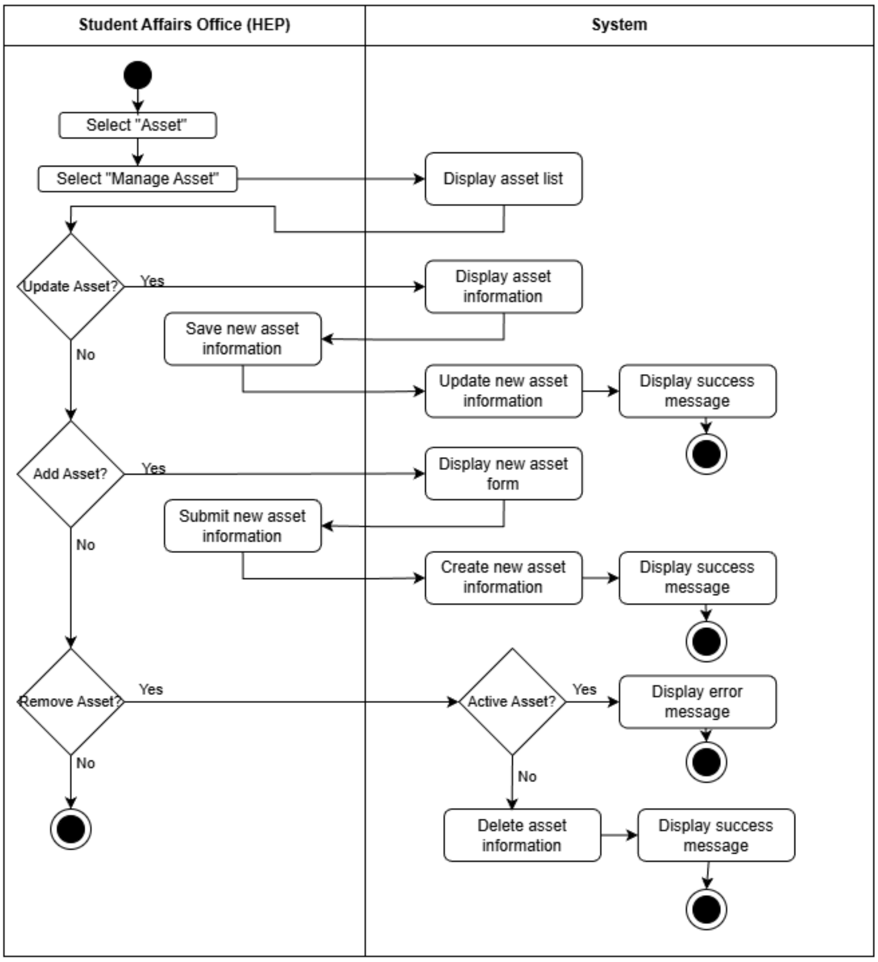

Figure 3.21.2 Activity Diagram for Manage Asset

4.6.2.22 US22 Issue Penalty

Table 4.22 shows the user story description, followed by the sequence diagram in Figure 3.22.1 and the activity diagram in Figure 3.22.2.

Table 4.22 User Story Description for Issue Penalty

User story: Issue Penalty
ID: US22
User Story Description: **As a** Student Affairs Office (HEP) **I want** to issue a penalty to students **So that** violations related to asset borrowing, such as late return or asset damages are properly managed.
Flow of events: 1. The Student Affairs Office logs into the system. 2. The Student Affairs Office selects the Asset function. 3. The Student Affairs Office selects the Returned Asset function. 4. The system displays a list of returned assets with detected violations sorted first. 5. The Student Affairs Office selects a return record. 6. The system displays application details and the violation, such as late return. 7. The Student Affairs Office reviews and determines the penalty type and penalty amount. 8. The Student Affairs Office confirms the penalty issuance. 9. The system records the penalty and updates the student's borrowing record.
Alternative flow: 1. The Student Affairs Office determines that no penalty is required. 2. The Student Affairs Office cancels the penalty issuance. 3. The system records the return as completed without penalty.
Acceptance Criteria: Precondition: 1. The Student Affairs Office user is authenticated and logged into the system. 2. The asset has been returned by the student. Postcondition: 1. The penalty is successfully recorded in the system.
Exception flow: 1. The entered penalty amount or type is invalid, the system prompts the Student Affairs Office to correct the information before confirmation

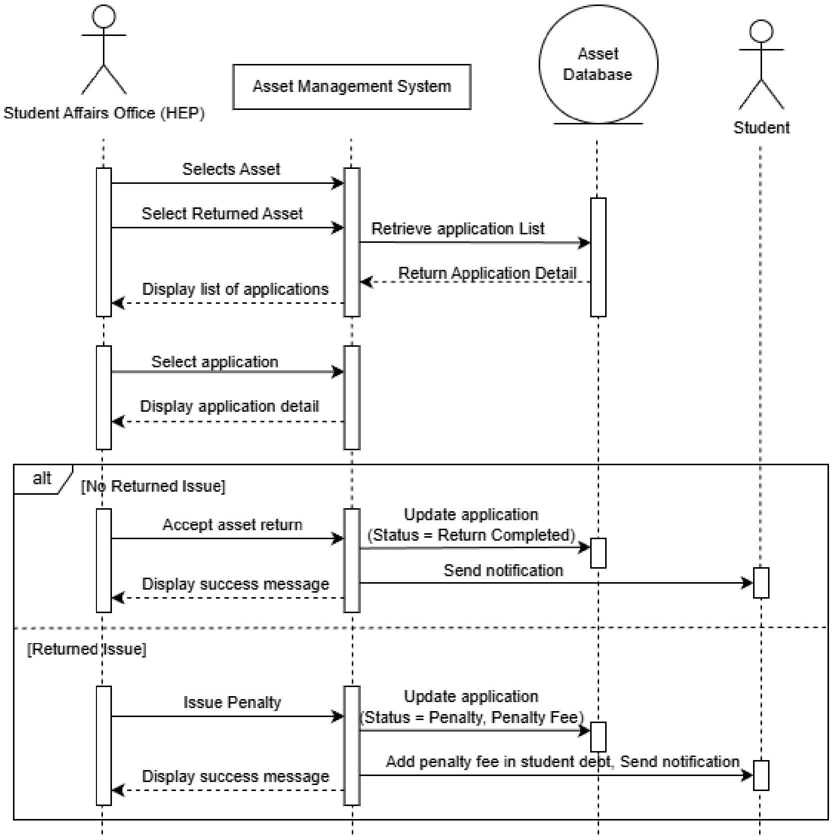

Figure 3.22.1 Sequence Diagram for Issue Penalty

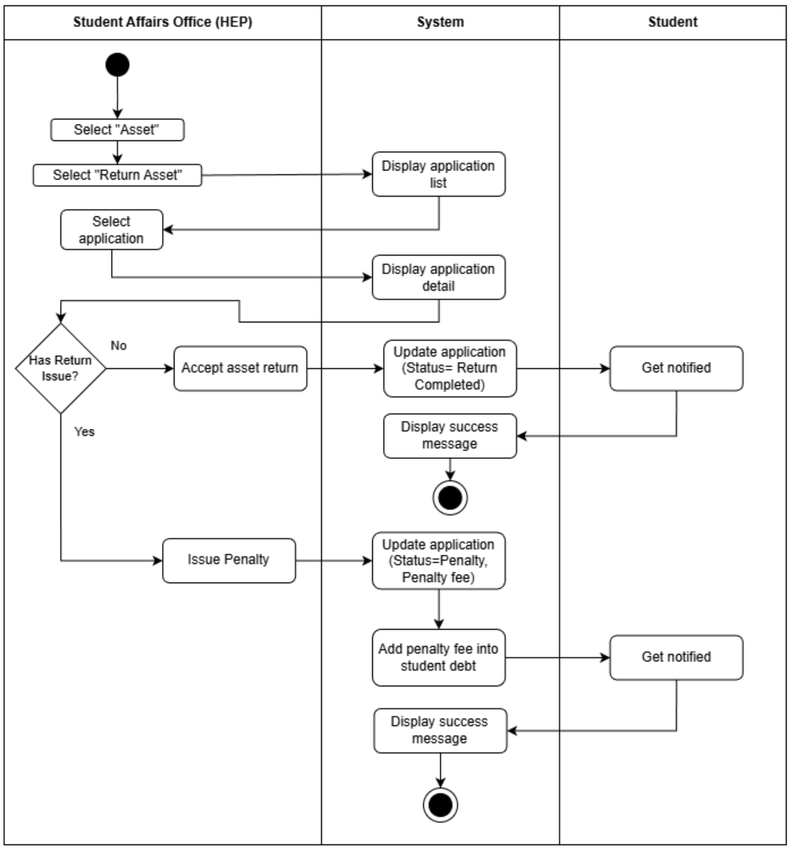

Figure 3.22.2 Activity Diagram for Issue Penalty

4.6.2.23 US23 Submit Financial Record

Table 4.23 shows the user story description, followed by the sequence diagram in Figure 3.23.1 and the activity diagram in Figure 3.23.2.

Table 4.23 User Story Description for Submit Financial Record

User story: Submit Financial Record
ID: US23
User Story Description: **As a** Club Committee **I want to** add and submit a financial record **So that** the club incomes and expenses can be recorded clearly to be tracked in the future
Flow of events: 1. The committee logs into the system. 2. The committee chooses the option to add financial records. 3. The committee selects the club or event, the amount to be added and the description. 4. The committee checks the information entered and submit the applications. 5. The system validates the financial record submission with proper event and amount. 6. The system displays the financial record in the club advisor' view for review purposes.
Alternative flow: 1. The system displays an error message if the club or event and the amount are missing.
Acceptance Criteria: Precondition: 1. The committee member is logged into the system using a verified club committee account. 2. The committee member has permission to submit financial records for the selected club or event. Postcondition: 1. A financial record with a positive amount (income) or negative amount (expense) is successfully submitted. 2. The submitted financial record is stored in the system. 3. The financial record is made available for the club advisor to review.
Exception flow: 1. If the committee enters an amount of zero, the system will display an error message. 2. If the committee attempts to submit a financial record for a club or event without proper authorization, an authorization error message is shown.

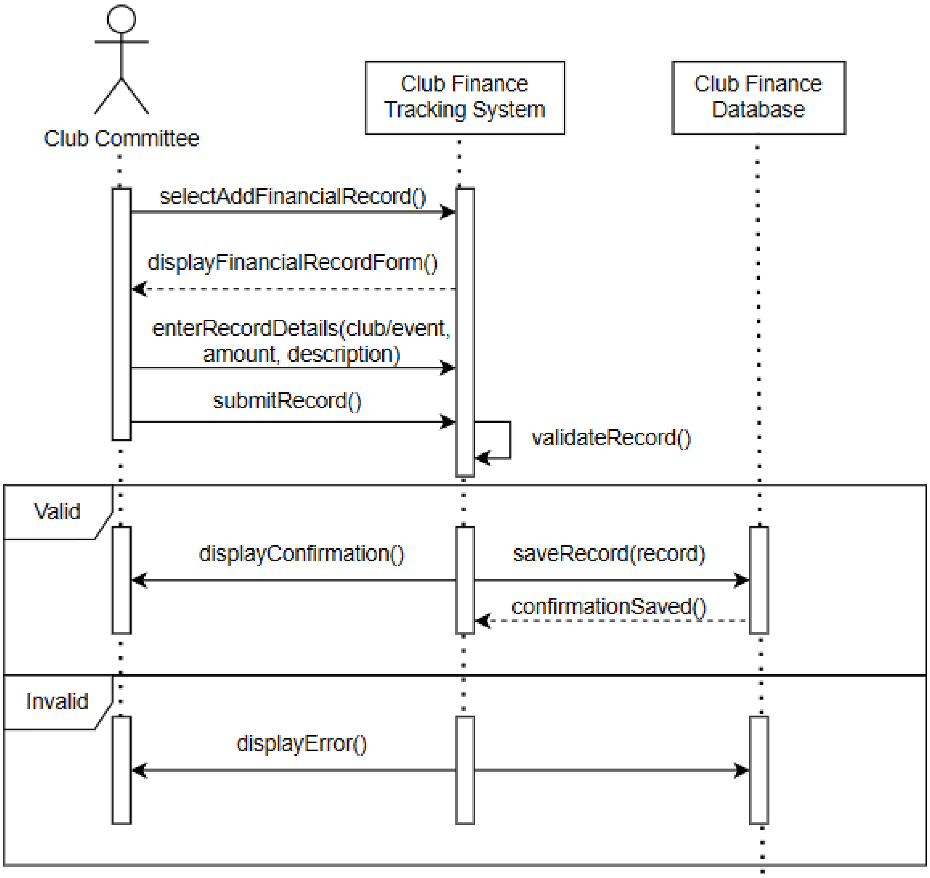

Figure 3.23.1 Sequence Diagram for Submit Financial Record

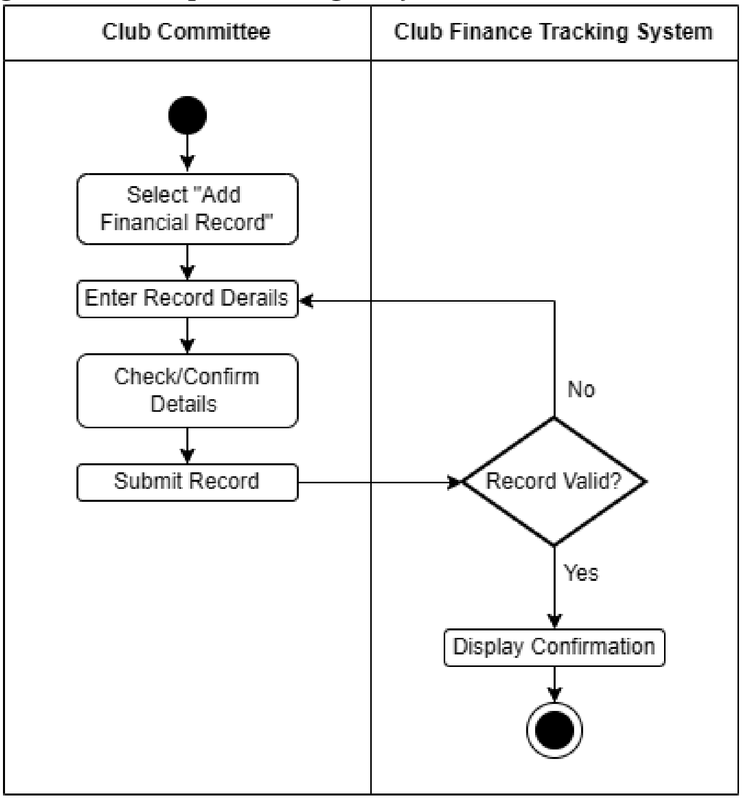

Figure 3.23.2 Activity Diagram for Submit Financial Record

4.6.2.24 US24 Verify Financial Record

Table 4.24 shows the user story description, followed by the sequence diagram in Figure 3.24.1 and the activity diagram in Figure 3.24.2.

Table 4.24 User Story Description for Verify Financial Record

User story: Verify Financial Record
ID: US24
User Story Description: **As a** Club Advisor **I want to** review and verify the financial record submitted by the club committee **So that** I can check and comment on the income and expenses of the club or event.
Flow of events: 1. The club advisor logged into the systems. 2. The club advisor selects the tab to review all financial records. 3. The system displays a list of submitted financial records. 4. The club advisor reviews the selected financial record. 5. The club advisor adds comments or notes if necessary. 6. The club advisor approves and verifies the financial records.
Alternative flow: 1. If no financial records are available, the system displays a message indicating that no records are found.
Acceptance Criteria: Precondition: 1. The club advisor is logged into the system using a valid advisor account. 2. At least one financial record has been submitted by the club committee. Postcondition: 1. The verify action and any comments are stored in the system.
Exception flow: If the advisor attempts to review without proper authorization, the system blocks access and displays an error message.

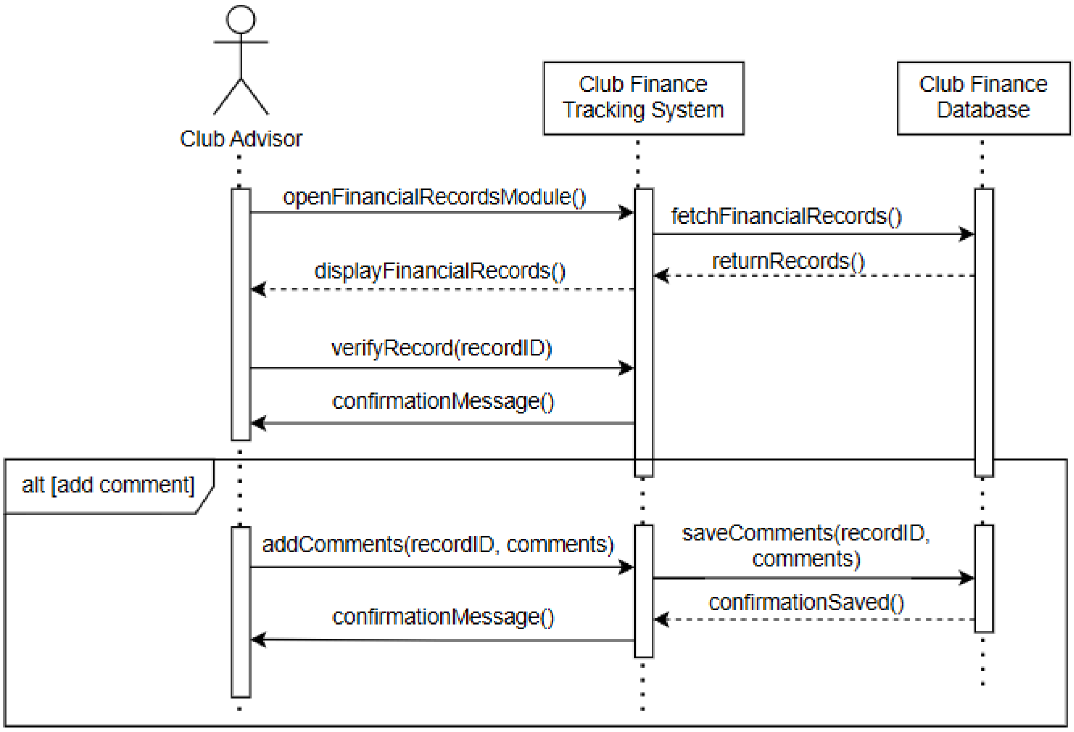

Figure 3.24.1 Sequence Diagram for Verify Financial Record

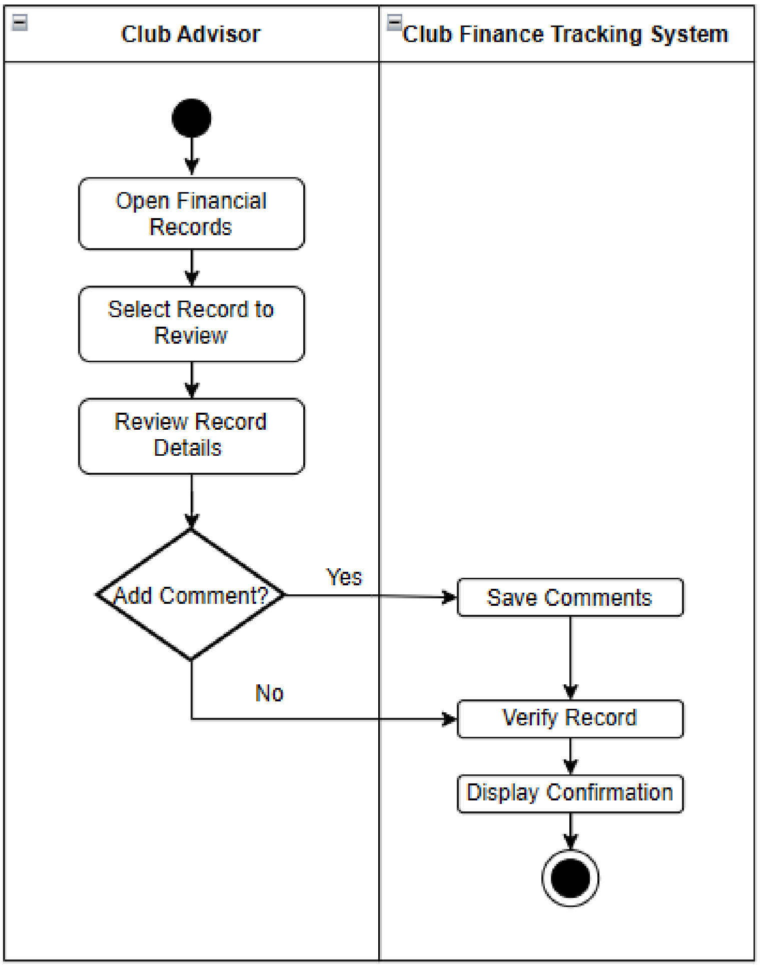

Figure 3.24.2 Activity Diagram for Verify Financial Record

4.6.2.25 View Financial Report

Table 4.25 shows the user story description, followed by the sequence diagram in Figure 3.25.1 and the activity diagram in Figure 3.25.2.

Table 4.25 User Story Description for View Financial Report

User story: View Financial Report
ID: US25
User Story Description: **As a** Student Affairs Officer **I want to** view club financial reports **So that** I can monitor club finances and access the submitted records for reference.
Flow of events: 1. The HEP officer logs into the system. 2. The officer selects the View Financial Reports option. 3. The system displays a list of all submitted financial records and summary reports. 4. The officer selects a report to view detailed financial data.
Alternative flow: 1. If there are no submitted financial records, the system displays a message indicating no data is available.
Acceptance Criteria: Precondition: 1. The HEP officer is logged into the system using a valid HEP account. 2. At least one financial record has been submitted by a club. Postcondition: 1. The officer can view financial records and reports successfully.
Exception flow: 1. If a non-HEP user attempts to access financial reports, the system denies access and displays an error message.

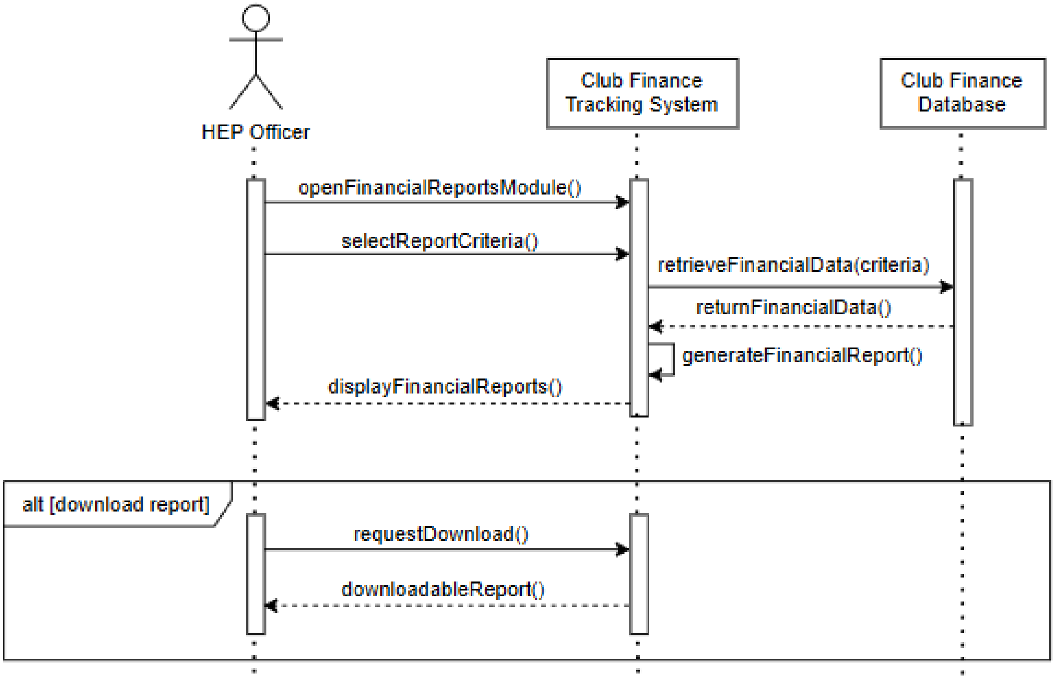

Figure 3.25.1 Sequence Diagram for View Financial Report

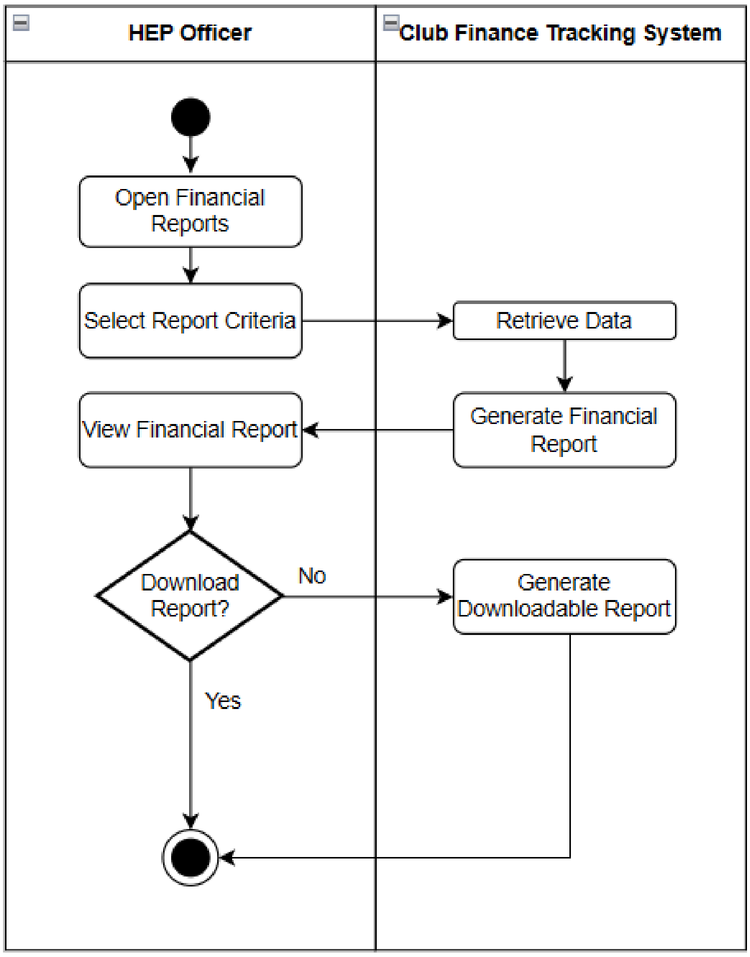

Figure 3.25.2 Activity Diagram for View Financial Report

4.6.3 Database Design

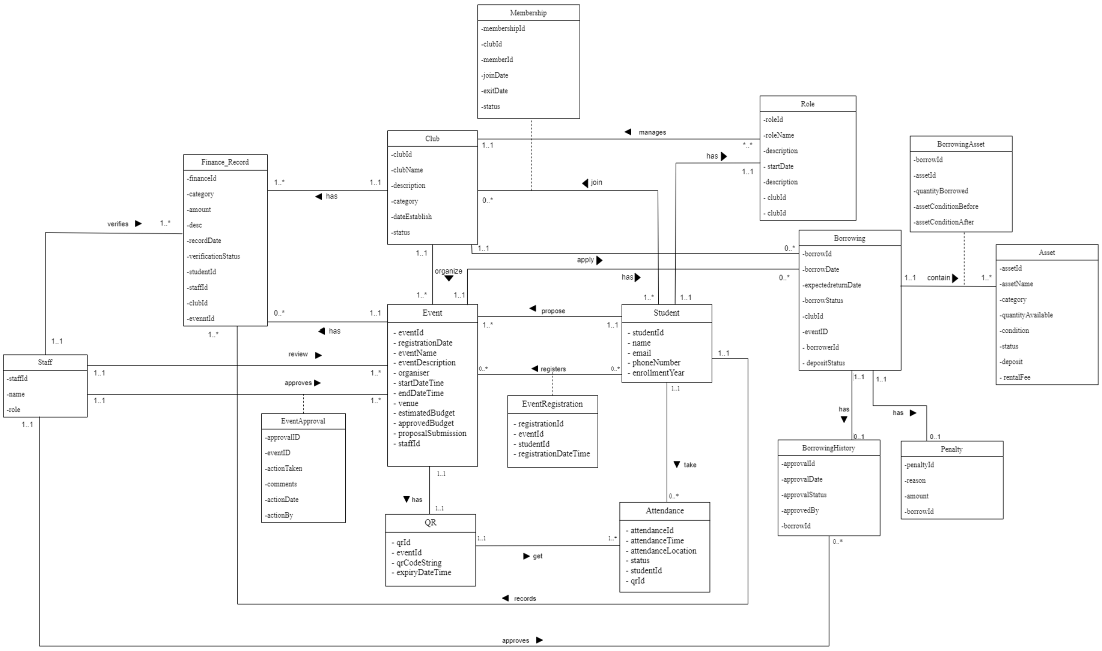

Figure 3.5.1 ERD

4.7 Summary

The background of the stakeholders is analysed, including the university departments associated with the proposed system. In addition, the existing system is examined to identify its limitations and the components that should be integrated into the proposed system. The system requirements are then defined, encompassing both functional and non-functional requirements, which will serve as the baseline for system development.

The use cases are identified and illustrated using a use case diagram. In summary, modules such as Club Profiles and Membership Management, Event Proposal and Approval Workflow, Event Registration and QR Code Generation, Event Attendance Scanning and Tracking, Asset Borrowing for Club Events, and Club Finance and Budget Tracking are identified for the Student Clubs and Activities Management System.

To further elaborate on the use cases, user stories are described and supported by sequence diagrams and activity diagrams. Based on the identified requirements, the database is designed and illustrated using an entity-relationship diagram.

CHAPTER 5
CONCLUSION

5.1 Introduction

This chapter concludes the Student Clubs and Activities Management System project. It discusses the limitations encountered during the development phase, highlights the overall contribution of the proposed system and makes recommendations for future improvements. The discussion in this chapter demonstrates how the proposed system advances the university's goal of building a more unified and efficient digital ecosystem while addressing the issue present in the current environment.

5.2 System Contribution

The system is designed to streamline processes and reduce manual procedures in the existing system by integrating multiple systems across different departments. It enables shared access to data sources and allows easier information retrieval based on users' roles and authorization levels.

The system adopts an enterprise architecture approach, which helps to unify and coordinate departmental processes across the university. Both staff and students benefit from the system. Staff users include but are not limited to academic staff, administrative staff and support staff. Student users include active students as well as students with various roles within the university.

5.3 System Constraints

The system is currently developed as a conceptual prototype, focusing on system design, documentation and user interface design using Figma. The actual implementation using SAP Build Apps, database and backend integration has not yet been fully implemented. Therefore, all features and functionality shown in this project are expected to be, not a fully functioning system.

In addition, the system uses sample data only and does not involve real student or staff data due to privacy and security considerations. Key features such as database connection, QR code scanning, user access control and integration with existing university systems (SMP and SMS) are only planned in the design and have not been developed yet. As a result, the system's performance, scalability and security in real-life use cannot be tested at this stage.

5.4 Future Suggestions

This system can be further enhanced with several improvements such as the implementation of offline support for attendance scanning. While an event is highly crowded, problems like unstable or limited internet connectivity could happen. Offline functionalities allow attendance data to be captured locally on device while automatically synchronized with the system database once connection is restored. This can significantly improve system reliability by reducing real-time dependency on network infrastructure.

In addition, the system can implement automatic backend operations of data archiving and lifecycle management activities. With students gradually graduating, old and unused event and attendance data can be automatically archived into a separate storage layer to maintain optimal system performance and database efficiency. This approach can ensure efficient use of storage resources while historical data can be preserved for future reference and analysis.

5.5 Summary

In order to improve the management of student activities across the university, this project recommended an enterprise architecture approach for the Student Clubs and Activities Management System. The system integrates several processes into a single platform to improve staff and student user experience and expand data accessibility. The system has a lot of potential to enable more efficient and well-coordinated university operations even though it is still simply a conceptual prototype. The suggested system can function as a reliable digital platform in line with the university's digital transformation initiatives with additional development and implementation.

Prototype Link:

https://www.figma.com/proto/M25ChRFutfHlhpEcCx9v1F/ERP--Clubs-and-Event-?node-id=1013-8&p=f&t=0dvb8PpfpQ0OCVCD-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1013%3A8&show-proto-sidebar=1

Video Link: https://youtu.be/yvmOK3xBE1c

GitHub Link: https://github.com/users/Yuylam/projects/2

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